Sensitivity Analysis in Sharp Regression Discontinuity Designs*

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Abstract

This paper develops a framework for extrapolating causal effects in sharp regression discontinuity (RD) designs beyond the local neighborhood of the cutoff. We adopt a partial identification approach that yields bounds on a broad class of causal parameters, including average and distributional treatment effects. Identification relies on smoothness restrictions on the distribution of potential outcomes with respect to the running variable, parameterized by sensitivity measures that capture the degree of smoothness and admit an interpretation in terms of statistical dependence. Within this framework, we derive sharp bounds on counterfactual distributions that can be estimated using standard nonparametric, semiparametric, or

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flexible parametric methods. The approach is well suited for sensitivity analysis, providing a transparent assessment of the robustness of extrapolated RD estimates. Monte Carlo simulations demonstrate its finite-sample performance. We also present an empirical application to illustrate how the proposed approach can be used to address empirical questions concerning external validity in sharp RD designs.

Keywords: regression discontinuity, sharp designs, extrapolation, external validity, partial identification, sensitivity analysis.
JEL Classification Numbers: C14, C31, C36, C51.

1 Introduction

Regression discontinuity (RD) designs are among the most widely used methods for causal inference with observational data. They apply to settings in which treatment assignment is determined by whether an observed score (i.e., the running variable) crosses a known cutoff value. Exploiting this feature, RD estimates are obtained by comparing the average outcome for observations located just above the cutoff with that for observations located just below the cutoff. Since their introduction by Thistlethwaite and Campbell (1960), RD designs have been increasingly used in the economics literature to estimate causal treatment effects.[1]

The seminal paper by Hahn et al. (2001) establishes conditions under which the RD estimate admits a causal meaning. The key condition is that the conditional expectation functions of the potential outcomes on the running variable are continuous at the cutoff value, which is referred to as "continuity assumption." Hahn et al. (2001) show that under the continuity assumption, the RD estimate is the average treatment effect (ATE) at the cutoff level in sharp RD designs when the treatment effects are heterogeneous. This identification result gives rise to an empirical challenge in that regression discontinuity estimates recover only local causal effects. The term "local" refers to treatment effects for a specific subpopulation—namely, units whose running variable values lie in a neighborhood of the cutoff—rather than effects for the entire population. Consequently, it is difficult to assess the external validity of regression discontinuity estimates, even though several empirical studies are explicitly concerned with this issue (e.g., Abdulkadiroğlu et al. (2014); Dobbie and Fryer Jr (2014); Hainmueller et al. (2015)).

This paper develops a framework for extrapolating causal effects in sharp regression discontinuity designs. The proposed approach enables identification of treatment effects for broader

[1] Lee and Lemieux (2010) and Cattaneo and Titiunik (2022) provide comprehensive surveys of RD designs.

subpopulations—and, under additional restrictions, potentially for the entire population—beyond the set of units at the cutoff. To achieve this, we adopt a partial identification strategy that delivers bounds on the causal parameters of interest. The partial identification approach allows us to learn about a broader class of causal parameters in sharp RD designs under weaker identifying assumptions, thereby yielding results that are more robust to deviations from point-identifying restrictions and improving the credibility of the analysis. As a byproduct, the framework naturally lends itself to sensitivity analysis.

Our key identifying assumption is a smoothness restriction. Specifically, we parameterize the degree of smoothness of the conditional distribution functions of the potential outcomes on the running variable. These parameters also serve as a sensitivity measure with respect to the identifying assumption. Importantly, they admit a natural interpretation in terms of the degree of statistical dependence between the potential outcomes and the running variable. These parameters are selected and varied by the researcher, thereby facilitating sensitivity analysis. Note that our identifying assumption is closely related to those introduced in Kline and Santos (2013), Kim et al. (2018), and Masten and Poirier (2018).

We derive the *sharp bounds* on the counterfactual distributions of the potential outcomes. These sharp bounds are tractable and easy to estimate in practice. Notably, the sharp bounds on the counterfactual distributions allow us to construct sharp bounds on functionals of these distributions, which serve as building blocks for identification of causal parameters of interest, such as the ATE and quantile treatment effects (QTEs), for broad subpopulations. In doing so, our framework can address the issue about external validity of RD estimates.

Since the resulting bounds may be wide, we introduce an additional identifying assumption that can help tighten them. Motivated by the stochastically monotone instrumental variable assumption of Mourifie et al. (2020), we propose the *stochastically monotone running variable assumption*. In an appropriate sense, this assumption characterizes the direction in which the conditional distribution functions of the potential outcomes change with the running variable as the conditioning value increases. As a consequence, it narrows the bounds on the counterfactual distributions of the potential outcomes and thereby yields tighter bounds on the causal parameters.

Based on the identification results, we conduct a small Monte Carlo simulation study and perform a sensitivity analysis. The simulation results demonstrate that our framework is effective for extrapolating regression discontinuity estimates, thereby allowing us to learn about a broader class of causal parameters in sharp RD designs.

We revisit the empirical study of Pop-Eleches and Urquiola (2013), who examine the effect of

attending better schools on students' academic performance, and use it to illustrate the empirical relevance of our approach. Our findings indicate that, with respect to the QTE, the RD estimates may not be representative of the relevant causal parameters. Specifically, over a wide range of quantile levels, the estimated bounds on the QTE conditional on values of the running variable away from the cutoff do not contain the RD estimates obtained at the cutoff. In addition, the estimated bounds on the unconditional QTE do not necessarily contain the RD estimates of the QTE at the cutoff. These findings suggest that, abstracting from sampling error, the QTE may vary across subpopulations indexed by the running variable, implying that the RD estimates at the cutoff need not be representative of the causal effect either for other subpopulations or for the population as a whole. In contrast, we do not find comparably clear evidence of such heterogeneity for the ATE.

We contribute to the literature in several ways. First, we develop a framework that enables extrapolation of regression discontinuity estimates to a broad class of causal parameters in sharp RD designs. By moving beyond treatment effects local to the cutoff, our framework directly addresses questions of external validity and enhances the policy relevance of RD estimates. In addition, we derive the *sharp* bounds on the conditional distributions of potential outcomes with respect to the running variable, which in turn enable construction of sharp bounds on distributional treatment effects in sharp regression discontinuity designs.

A growing body of literature has examined related concerns, and our approach contributes to this line of research by providing a partial identification approach. Angrist and Rokkanen (2015) propose an approach to extrapolation of RD estimates that relies on a conditional independence assumption.[2] Their approach requires the existence of a set of covariates such that the running variable is conditionally independent of the potential outcomes given these covariates. Cattaneo et al. (2021) develop a method for extrapolating regression discontinuity estimates based on multiple cutoffs, which shares conceptual similarities with difference-in-differences designs. Deaner and Kwon (2025) exploit a comonotonicity assumption to extrapolate the conditional average treatment effect away from the cutoff, imposing conditional monotonicity on the conditional expectation functions of the potential outcomes with respect to the running variable. Feng and Ruggieri (2025) propose a structural modeling framework for extrapolating average partial effects in regression discontinuity designs. Our approach does not require additional variables or structural modeling and imposes no distributional assumptions. As a result, we complement these studies by adopting a partial

[2] Rokkanen (2015) also considers a similar approach to extrapolation of RD estimates in sharp designs.

identification strategy based on a parsimonious modeling framework.[3]

Second, we derive identification results for a broader class of causal parameters than those considered in prior studies. All of the aforementioned studies have focused primarily on the extrapolation of the average RD treatment effect. In contrast, we provide results on the extrapolation of not only the average RD treatment effect but also distributional RD treatment effects. Consequently, our results allow researchers to investigate heterogeneity in treatment effects, in addition to the extrapolation of RD treatment effects. Sun (2023) adopts a partial identification approach under an assumption similar to ours, but her analysis focuses primarily on the extrapolation of average treatment effects in RD designs with potentially multiple cutoff values.

Third, this paper contributes to the literature on sensitivity analysis. Sensitivity analysis is valuable for empirical researchers seeking to assess the robustness of empirical conclusions to variations in identifying assumptions, data, or modeling choices. Our framework builds primarily on the approaches developed by Kline and Santos (2013) and Masten and Poirier (2018), who propose sensitivity analysis frameworks in their respective settings.[4]

The rest of this paper is organized as follows. In Section 2, we introduce the model and identifying assumptions. Section 3 provides the identification results and formalizes the idea of the stochastically monotone running variable assumption. Section 4 briefly discusses the estimation and inference for the bounds on some causal parameters, and section 5 presents the Monte Carlo simulation results. Section 6 provides an empirical illustration, and Section 7 concludes. All mathematical proofs are provided in the Appendix.

2 Model and Assumptions

Let $(\Omega, \mathcal{S}, \mathbf{P})$ be a probability space. Let R denote the running variable and c be the cutoff level. Then, in a sharp RD design, $D = \mathbf{1}(R \geq c)$, where D is the observed binary treatment variable and $\mathbf{1}(\cdot)$ is an indicator function. Letting Y_1 and Y_0 be the potential outcomes, the observed outcome variable Y is $Y = DY_1 + (1 - D)Y_0$.

Let h be a measurable function such that $E[|h(Y_d)|] < \infty$ for all $d \in \{0, 1\}$ For a generic set A, $\bar{A}$ is the closure A. We denote the support of a random variable W by $Supp(W)$ and the distribution

[3] Although we focus on extrapolation in sharp RD designs, it is worth noting that a number of papers in the literature address extrapolation in fuzzy RD designs, including Battistin and Rettore (2008), Mealli and Rampichini (2012), and Bertanha and Imbens (2020).

[4] There are a growing number of papers that consider sensitivity analysis, including Kline and Santos (2013), Manski and Pepper (2018), Masten and Poirier (2018), Masten and Poirier (2020), Rambachan and Roth (2023), and Masten et al. (2024), among others.

function by $F_W(\cdot)$.[5] We define the "treated region" and "control region" as $\mathcal{R}_1 \equiv [c, \infty) \cap Supp(R)$ and $\mathcal{R}_0 \equiv (-\infty, c) \cap Supp(R)$, respectively.

Assumption 2.1. *The sets $\mathcal{R}_1$ and $\mathcal{R}_0$ are nonempty, and for any $r_1 \in \mathcal{R}_1$ and $r_0 \in \mathcal{R}_0$,*

$$E\left[h\left(Y_1\right) \mid R = r_1\right] = E\left[h\left(Y\right) \mid R = r_1\right]$$

and

$$E\left[h\left(Y_0\right) \mid R = r_0\right] = E\left[h\left(Y\right) \mid R = r_0\right].$$

Assumption 2.1 requires that $E\left[h\left(Y_1\right)|R_1 = r_1\right]$ and $E\left[h\left(Y_0\right) \mid R = r_0\right]$ are identified from the distribution of (Y, D, R) and provides baseline estimators for extrapolation of treatment effects. Specifically, for given $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$, we use $E\left[h(Y_1) \mid R = r_1\right]$ to learn about $E\left[h(Y_1) \mid R = r_0\right]$ and use $E\left[h(Y_0) \mid R = r_0\right]$ to learn about $E\left[h(Y_0) \mid R = r_1\right]$. When $c \in Supp(R)$, the treatment effect at the cutoff can be identified under continuity assumptions (cf. Hahn et al. (2001)). It is also worth pointing out that $r_1 \geq r_0$ for any $r_d \in \mathcal{R}_d$ with $d \in \{0, 1\}$.

The next assumption is our main assumption for the extrapolation of treatment effects in sharp RD designs.

Assumption 2.2. *There exist $k_{0,h}, k_{1,h} \geq 0$ such that for all $r, \tilde{r} \in Supp(R)$,*

$$\left|E\left[h\left(Y_1\right)|R = r\right] - E\left[h\left(Y_1\right)|R = \tilde{r}\right]\right| \leq k_{1,h}\left|r - \tilde{r}\right|$$

and

$$\left|E\left[h\left(Y_0\right)|R = r\right] - E\left[h\left(Y_0\right)|R = \tilde{r}\right]\right| \leq k_{0,h}\left|r - \tilde{r}\right|.$$

Assumption 2.2 is a smoothness condition imposed on the conditional expectation functions.When $c \in Supp(R)$ and $h(y) = y$, it is stronger than the continuity-at-the-cutoff condition in Hahn et al. (2001) (i.e., Assumption (A1) in Hahn et al. (2001)). While Assumption (A1) in Hahn et al. (2001) only requires continuity at $R = c$, Assumption 2.2 requires $E[h(Y_d) \mid R = r]$ to be globally Lipschitz continuous in r for $d \in \{0, 1\}$.

The constants $k_{0,h}$ and $k_{1,h}$ can be interpreted as sensitivity parameters for smoothness, as in Kim et al. (2018). They may also be interpreted as capturing the degree of mean independence

[5] By definition, $Supp(W)$ is the smallest closed set outside of which $F_W(\cdot)$ is either 0 or 1 (cf. Folland (1999, p.132)).

between $h(Y_d)$ and R. Given the generality of h, these constants can further be understood as measuring the degree of independence between the potential outcomes and the running variable. In particular, when $k_{d,h} = 0$ for a given $d \in \{0,1\}$ and $h(Y) = h_y(Y) \equiv \mathbf{1}(Y \leq y)$ for $y \in \mathbb{R}$, Assumption 2.2 implies that Y_d is independent of R, as in Kline and Santos (2013) and Masten and Poirier (2018). Note that these constants are chosen by researchers.

There are several papers in the literature that consider restrictions similar to those in Assumption 2.2. For example, Kim et al. (2018) introduce smooth treatment response and smooth monotone treatment response assumptions of Lipschitz continuity of the conditional expectation functions of the potential outcomes with respect to the treatment. In addition, Manski and Pepper (2018) consider similar assumptions that impose bounded variation on potential outcomes with respect to treatments. Assumption 2.2 differs from them in that the smoothness assumption is imposed with respect to the conditioning variable, namely the running variable.[6]

3 Identification

The following theorem provides bounds on the conditional expectation functions that will serve as the building blocks for constructing sharp identified regions for a certain causal parameter of interest.

Theorem 3.1. *Suppose that Assumptions 2.1 and 2.2 hold. Let h be a measurable function such that $E[|h(Y_d)|] < \infty$ for all $d \in \{0,1\}$. Then, for any $r \in Supp(R)$, we have*

$$E[h(Y_1)|R = r] \in [LB_1(k_{1,h}, r), UB_1(k_{1,h}, r)],$$
$$E[h(Y_0)|R = r] \in [LB_0(k_{0,h}, r), UB_0(k_{0,h}, r)],$$

[6] A related assumption in the literature is the monotone treatment selection (MTS) assumption introduced by Manski and Pepper (2000). However, the MTS assumption imposes monotonicity on the conditional expectation of the potential outcome with respect to the counterfactual treatment variable, which differs from Assumption 2.2. In addition, the monotone instrumental variables (MIV) assumption in Manski and Pepper (2000) is also closely related to Assumption 2.2. Nevertheless, unlike MTS or MIV, we do not impose monotonicity on the conditional expectation functions. Instead, we restrict the range within which the counterfactual conditional expectation functions can lie.

where

$$
\begin{aligned}
LB_1(k_{1,h}, r) &\equiv \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E\left[h\left(Y_1\right) | R = \tilde{r}\right] - k_{1,h} \cdot \left| r - \tilde{r} \right| \right\}, \\
UB_1(k_{1,h}, r) &\equiv \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E\left[h\left(Y_1\right) | R = \tilde{r}\right] + k_{1,h} \cdot \left| r - \tilde{r} \right| \right\}, \\
LB_0(k_{0,h}, r) &\equiv \sup_{\tilde{r} \in \mathcal{R}_0} \left\{ E\left[h\left(Y_0\right) | R = \tilde{r}\right] - k_{0,h} \cdot \left| \tilde{r} - r \right| \right\}, \\
UB_0(k_{0,h}, r) &\equiv \inf_{\tilde{r} \in \mathcal{R}_0} \left\{ E\left[h\left(Y_0\right) | R = \tilde{r}\right] + k_{0,h} \cdot \left| \tilde{r} - r \right| \right\}.
\end{aligned}
$$

The bounds in Theorem 3.1 have a similar structure to that of Manski's bounds under the monotone instrumental variables bounds (cf. Manski and Pepper (2000)) and bounds on the counterfactual mean function in Kim et al. (2018).

We can construct bounds on the conditional expectation functions over regions in which they may not be point identified. Suppose, for example, that $Supp(R) = [r_{min}, r_{max}]$ for some constants r_{min} and r_{max} with $r_{min} < c < r_{max}$ and that $\mathcal{R}_1 = [c, r_{max}]$ and $\mathcal{R}_0 = [r_{min}, c)$. Focusing on the case that $h(y) = y$, it is evident that the counterfactual mean $E\left[Y_d | R = r\right]$ is not identified for $r \in \mathcal{R}_{1-d}$ without further restrictions. Nevertheless, we can partially identify $E\left[Y_d | R = r\right]$ for $r \in \mathcal{R}_{1-d}$ by using the results in Theorem 3.1.

The bounds can be constructed by replacing the conditional expectation functions with their consistent estimators—whether parametric, semiparametric, or nonparametric—and then taking the supremum and/or infimum over the relevant region. However, in sharp RD designs, these intersection bounds in Theorem 3.1 can be further reduced. To this end, let $r_1^\star \equiv \inf \mathcal{R}_1$ and $r_0^\star \equiv \sup \mathcal{R}_0$.

Theorem 3.2. *Suppose that Assumptions 2.1 and 2.2 hold. Let $r_d \in \mathcal{R}_d$ for each $d \in \{0, 1\}$. Then, we have*

$$
\begin{aligned}
LB_1(k_{1,h}, r_0) &= E\left[h\left(Y_1\right) | R = r_1^\star\right] - k_{1,h} \cdot \left| r_0 - r_1^\star \right|, \\
UB_1(k_{1,h}, r_0) &= E\left[h\left(Y_1\right) | R = r_1^\star\right] + k_{1,h} \cdot \left| r_0 - r_1^\star \right|, \\
LB_0(k_{0,h}, r_1) &= E\left[h\left(Y_0\right) | R = r_0^\star\right] - k_{0,h} \cdot \left| r_1 - r_0^\star \right|, \\
UB_0(k_{0,h}, r_1) &= E\left[h\left(Y_0\right) | R = r_0^\star\right] + k_{0,h} \cdot \left| r_1 - r_0^\star \right|.
\end{aligned}
$$

Theorem 3.2 allows us to avoid searching over $\mathcal{R}_1$ and $\mathcal{R}_0$ to compute the supremum and infimum in the intersection bounds in Theorem 3.1. the argsup and arginf to construct the bounds.

To illustrate the empirical relevance of the results, suppose that $Supp(R) = [r_{min}, r_{max}]$ for some constants r_{min} and r_{max} with $r_{min} < c < r_{max}$ and that $\mathcal{R}_1 = [c, r_{max}]$ and $\mathcal{R}_0 = [r_{min}, c)$, and therefore, $r_1^\star = r_0^\star = c$. In this case, Theorem 3.2 implies that estimating the bounds requires only the estimation of $E[h(Y_1)|R = c] = E[h(Y)|R = c]$ and $E[h(Y_0)|R = c] = \lim_{r \uparrow c} E[h(Y)|R = r]$, which can be implemented using the well-known nonparametric approach proposed by Calonico et al. (2014).

The intuition behind the results in Theorem 3.2 is based on the geometric fact that, for a fixed $r_0 \in \mathcal{R}_0$ and for any $\tilde{r} \in \mathcal{R}_1$, $r_1^\star$ is the unique point such that $|r_0 - \tilde{r}| = |r_0 - r_1^\star| + |\tilde{r} - r_1^\star|$. Moving from $r_1^\star$ to an interior point of $\mathcal{R}_1$ increases the distance to r_0 by $|\tilde{r} - r_1^\star|$. Under Assumption 2.2, the maximal admissible change in the value of $g_1(\tilde{r})$ from $g_1(r_1^\star)$ is $k_{1,h}\left|\tilde{r} - r_1^\star\right|$.

Figure 3.1 illustrates this intuition. The blue solid line in Figure 3.1 represents $g_1(r) = E[Y_1|R = r]$ over $\mathcal{R}_1$ (i.e., $h(x) = x$)). Our objective is to identify $E[Y_1|R = r_0]$ for a given $r_0 \in \mathcal{R}_0$, which is the target point.The dashed lines correspond to the upper and lower bounds on $E[Y_1|R = r_0]$, constructed by using $E[Y_1|R = r_1]$ for $r_1 \in \mathcal{R}_1$, which are indicated by the circles on the blue solid line. As r_1 moves farther away from r_0, the bounds widen as the $k_{1,h}|r_1 - r_0|$ increases. Focusing on the lower bound, recall that under Assumption 2.2, for a fixed $r_0 \in \mathcal{R}_0$,

$$E[Y_1|R = r_0] \geq E[Y_1|R = r_1] - k_{1,h} \cdot \left|r_0 - r_1\right|$$

for $r_1 \in \mathcal{R}_1$. Even when $E[Y_1|R = r]$ is assumed to be increasing in r, the potential gain from choosing a larger r_1, which would make $E[Y_1|R = r_1]$ closer to $E[Y_1|R = r_0]$, is dominated by the term $k_{1,h} \cdot \left|r_0 - r_1\right|$ under Assumption 2.2. Consequently, the bound is tightened only by selecting $r_1 \in \mathcal{R}_1$ as close as possible to r_0. Owing to the sharp RD feature, the closet point to r_0 in $\mathcal{R}_1$ is $r_1^\star$.

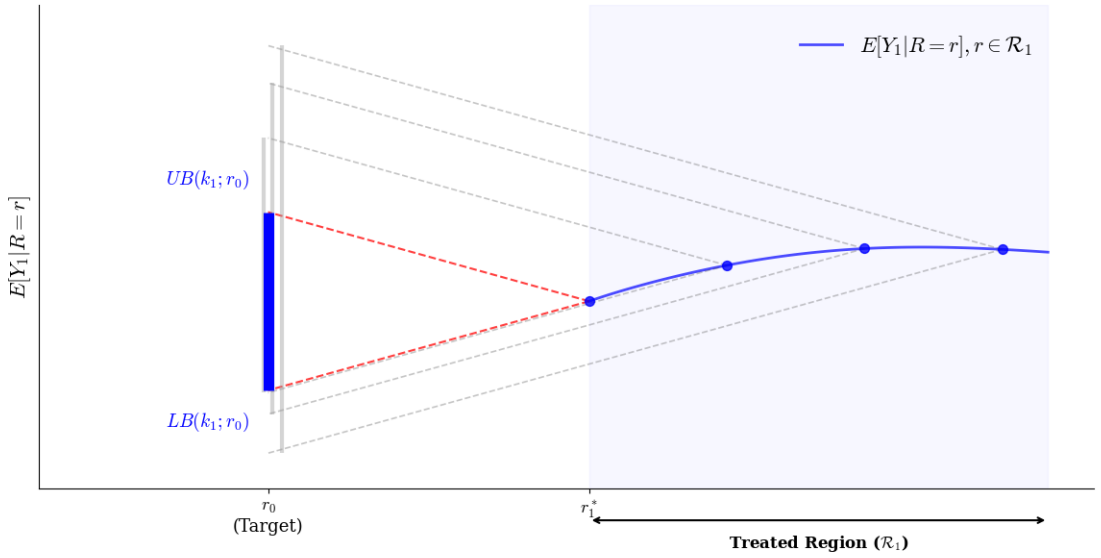
It is worth pointing out that $r_d^\star \in \mathcal{R}_d$ is not required for the results in Theorem 3.2 to hold and it is enough to assume that $r_d^\star \in Supp(R)$. If $r_d^\star \notin \mathcal{R}_d$, $E[h(Y_d)|R = r_d^\star]$ can be understood as the one-side limit of the corresponding conditional mean function, $\lim_{r \to r_d^\star : r \in \mathcal{R}_d} E[h(Y_d)|R = r]$. In addition, Theorem 3.2 facilitates the proof of sharpness for some specific choice of h.[7]

3.1 Identification of the Distributions of Counterfactual Outcomes

We now turn to the identification of the distributions of counterfactual outcomes. Identification of distribution functions is important because it enables the analysis of distributional treatment

[7] See the proof of Theorem 3.3.

Figure 3.1: Extrapolation at the Boundary



effects, such as quantile treatment effects, and facilitates the study of heterogeneous treatment effects across the distribution of the outcome variable of interest. Several papers in the literature examine distributional effects in regression discontinuity designs, including Frandsen et al. (2012), Shen and Zhang (2016), and Qu and Yoon (2019) just to name a few. Building on this literature, we extend our identification results for conditional mean functions to the distributions of potential outcomes. For generic random variables U and V, we denote the conditional distribution of U on $V = v$ by $F_{U|V}(\cdot|v)$.

Assumption 3.1. *The following conditions hold:*

(i) For each $d \in \{0, 1\}$ and for all $y \in \mathbb{R}$, $F_{Y_d|R}(y|r_d)$ is identified for any $r_d \in \mathcal{R}_d$.

(ii) There exist $k_{0,c}, k_{1,c} \geq 0$ such that for all $r, \tilde{r} \in Supp(R)$,

$$\begin{aligned} \sup_{y \in \mathbb{R}} \left| F_{Y_1|R}(y|r) - F_{Y_1|R}(y|\tilde{r}) \right| &\leq k_{1,c} \left| r - \tilde{r} \right|, \\ \sup_{y \in \mathbb{R}} \left| F_{Y_0|R}(y|r) - F_{Y_0|R}(y|\tilde{r}) \right| &\leq k_{0,c} \left| r - \tilde{r} \right|. \end{aligned} \tag{1}$$

Assumption 3.1-(i) requires that we can identify the conditional distributions of counterfactual outcomes for some subpopulations. Frandsen et al. (2012) and Shen and Zhang (2016) consider

the identification of conditional distributions of the potential outcomes on the running variable at the cutoff point. Similar to Hahn et al. (2001), the key identifying assumption is the continuity assumption at the cutoff value. Specifically, they require that for all $y \in \mathbb{R}$ and $d \in \{0, 1\}$, $F_{Y_d|R}(y|r)$ is continuous at $r = c$. Under this assumption, they show that distributional causal parameters can be identified.

Motivated by Kline and Santos (2013) and Masten and Poirier (2018), Assumption 3.1-(ii) imposes a condition that the distance between two distributions is small when the conditioning values are close. Specifically, the left hand sides of the equations in Assumption 3.1-(ii) are the Kolmogorov-Smirnov distance between two distributions, whereas the right hand sides are distances between r and $\tilde{r}$. Related to this point, Assumption 3.1-(ii) can be interpreted as the continuity assumption on the distribution functions, which strengthens the continuity assumption in Frandsen et al. (2012) and Shen and Zhang (2016) in the sense that the distribution functions are continuous on $Supp(R)$.

Moreover, Assumption 3.1-(ii) characterizes the statistical dependence between the potential outcomes and running variable. When $k_{d,c} = 0$, the potential outcome Y_d is statistically independent of the running variable R. As the value of $k_{d,c}$ increases, a greater degree of dependence between Y_d and MR is allowed. Consistent with this interpretation, a number of studies in the literature adopt assumptions that are similar in spirit to Assumption 3.1-(ii). Kline and Santos (2013) consider a restriction on the distance between the distributions of the outcome variable for selected and non-selected units, which captures the degree to which a missing-at-random assumption holds in a sample selection framework. Masten and Poirier (2018) introduce the concept of c-independence, which parameterizes the degree of conditional independence in a potential outcomes model with a binary treatment. Although Rambachan and Roth (2023) do not directly characterize the degree of dependence between variables, they propose a related identifying assumption that quantifies the degree of violation of the parallel trends assumption in a difference-in-differences framework through a nuisance parameter analogous to $k_{d,c}$ in this paper.

It is important to note that the choice of $k_{d,c}$ for $d \in \{0, 1\}$ can affect the identifying power of Assumption 3.1-(ii). Since the left hand sides in (1) are bounded by 1, there is no restriction imposed on the distributions when the right hand sides are greater than or equal to 1. Related to this point, we point out that it is possible to rescale the running variable before choosing the value of $k_{d,c}$. Specifically, for any affine transformation of R, $R' = aR + b$ with $a \neq 0$ and $b \in \mathbb{R}$, Assumption 3.1-(ii) can be rewritten as $\sup_{y \in \mathbb{R}} |F_{Y_d|R'}(y \mid r') - F_{Y_d|R'}(y \mid \tilde{r}')| \leq k'_{d,c} \left| r' - \tilde{r}' \right|$ with

$k_{d,c}^{'} = k_{d,c}/|a|$, while the right hand side is $k_{d,c}^{'}\left|r^{'} - \tilde{r}^{'}\right| = \frac{k_{d,c}}{|a|}|a|\left|r - \tilde{r}\right| = k_{d,c}\left|r - \tilde{r}\right|$. This implies that the condition $k_{d,c}|r - \tilde{r}| < 1$ is invariant to an affine transformation of R with the corresponding rescaling of $k_{d,c}$.

Theorem 3.3. *Suppose that Assumption 3.1 holds. Let $r_d \in \mathcal{R}_d$ for each $d \in \{0,1\}$ be given and $d^{'} \equiv 1 - d$. Define*

$$\begin{aligned} \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) &\equiv \max\left\{F_{Y_d|R}(y \mid r_d^\star) - k_{d,c}|r_{d'} - r_d^\star|, 0\right\}, \\ \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) &\equiv \min\left\{F_{Y_d|R}(y \mid r_d^\star) + k_{d,c}|r_{d'} - r_d^\star|, 1\right\}. \end{aligned} \tag{2}$$

Then, we have

(a) $F_{Y_d|R}(\cdot \mid r_{d'}) \in \mathcal{F}^{k_{d,c}}_{Y_d|r_{d'}}$, where

$$\mathcal{F}^{k_{d,c}}_{Y_d|r_{d'}} \equiv \left\{F \in \mathcal{F}_{\mathbb{R}} : \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \le F(y) \le \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}), \forall y \in \mathbb{R}\right\};$$

(b) For each $d \in \{0,1\}$ and for each $\varepsilon_d \in [0,1]$ and $\eta_d \in (0, \min\{\delta_d, (1-\delta_d)/2\})$, there exists a cdf $F^{k_d}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}^{k_d}_{Y_d|r_{d'}}$ such that

(i) for any $r_0 \in \mathcal{R}_0$ and $r_1 \in \mathcal{R}_1$, there exists a joint distribution of (Y_1, Y_0, R, D) consistent with our maintained assumptions such that, for all $y \in \mathbb{R}$,

$$\begin{aligned} \Pr(Y_1 \le y \mid R = r_0) &= F^{k_{1,c}}_{Y_1|R}(y \mid r_0; \varepsilon_1, \eta_1), \\ \Pr(Y_0 \le y \mid R = r_1) &= F^{k_{0,c}}_{Y_0|R}(y \mid r_1; \varepsilon_0, \eta_0); \end{aligned}$$

(ii) for each $d \in \{0,1\}$ and each $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid r_d^\star) < 1$, as $\eta_d \searrow 0$,

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 1, \eta_d) \nearrow \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'})$$

and

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 0, \eta_d) \searrow \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'});$$

(iii) for each $y \in \mathbb{R}$ and $r_{d'} \in \mathcal{R}_{d'}$, the function $\varepsilon_d \mapsto F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous on $[0,1]$;

(c) Consequently, for any $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid r_d^\star) < 1$, the pointwise bounds

$$F_{Y_d|R}(y \mid r_{d'}) \in \left[\underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}), \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'})\right]$$

have a sharp interior.

Theorem 3.3 establishes the sharp bounds on the conditional distribution functions of the potential outcomes on the running variable, which are useful for extrapolation of treatment effects. The proof of Theorem 3.3 builds on the proof of Proposition 2 in Masten and Poirier (2018), but there is a substantial difference between the two. Importantly, while Masten and Poirier (2018) consider the c-independence assumption in a binary treatment effect model—where the conditioning variable is binary—our results generalize their framework to settings in which the conditioning variable (i.e., the running variable) may be either discrete or continuous. As a result, we cannot directly apply the results of Masten and Poirier (2018), since this generalization renders the establishment of sharpness of the bounds more technically challenging.

The bounds in (2) are sharp in the sense that, for $d \in \{0,1\}$ and $d' = 1-d$, there exist attainable distribution functions $F^{k_{d,c}}_{Y_d|R}(\cdot|r_{d'};\epsilon_d,\eta_d)$ that can be made arbitrarily close to the lower and upper bounds and that vary continuously between these bound functions, as pointed out by Masten and Poirier (2018). These properties of the bounds are essential for constructing sharp identified regions for functionals of the conditional distribution functions that correspond to causal parameters of interest, such as (conditional) average or quantile treatment effects.

The identification result presented in Theorem (3.3) allows us to construct bounds on the unconditional distributions of the potential outcomes. Specifically, since the distribution of R, F_R, is identified from the data, it is possible to derive bounds on the unconditional distribution function of Y_d by using the fact that

$$F_{Y_d}(y) = \int F_{Y_d|R}(y|r)dF_R(r).$$

The following corollary provides the identification result for the unconditional distribution functions of the potential outcomes.

Corollary 3.4. *Suppose that Assumptions 3.1 and 3.2 hold. Let $d \in \{0,1\}$ be given and $d' = 1-d$. Then, for each $y \in \mathbb{R}$,*

$$F_{Y_d}(y) \in \left[\int_{\mathcal{R}_{d'}} \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r)dF_R(r) + \int_{\mathcal{R}_d} F_{Y|R}(y|r)dF_R(r), \int_{\mathcal{R}_{d'}} \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r)dF_R(r) + \int_{\mathcal{R}_d} F_{Y|R}(y|r)dF_R(r)\right].$$

3.2 Identification of Causal Parameters

We consider the identification of causal parameters, such as the conditional quantile treatment effect (CQTE) and conditional average treatment effect (CATE).

For each $d \in \{0,1\}$ and $\tau \in (0,1)$, define

$$Q_{Y_d|R}(\tau \mid r) \equiv \inf\left\{y \in Supp(Y_d \mid R = r) : F_{Y_d|R}(y \mid r) \geq \tau\right\}$$

as the τ-th conditional quantile function of the potential outcome Y_d on $R = r$. Then the τ-th CQTE given $R = r$ is

$$CQTE(\tau|r) \equiv Q_{Y_1|R}(\tau|r) - Q_{Y_0|R}(\tau|r).$$

From the CQTE, we can identify the CATE on $R = r$ as

$$CATE(r) \equiv E\left[Y_1 - Y_0|R = r\right] = \int_0^1 CQTE(\tau|r)d\tau.$$

We provide closed-form bounds on the quantile function of the potential outcome Y_d given $R = r_{d'}$, where for each $d \in \{0,1\}$, $d^{'} = 1 - d$ and $r_{d'} \in \mathcal{R}_{d'}$. Let

$$\delta_d \equiv \delta_d(r_{d'}) \equiv k_{d,c}|r_{d'} - r_d^\star|.$$

For $\tau \in (0,1)$ and $r_{d'} \in \mathcal{R}_{d'}$, define

$$\begin{aligned}
\underline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'}) &:= \begin{cases} -\infty & \tau \leq \delta_d, \\ Q_{Y|R,D=d}(\tau - \delta_d \mid r_d^\star) & \tau > \delta_d, \end{cases} \\
\overline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'}) &:= \begin{cases} Q_{Y|R,D=d}(\tau + \delta_d \mid r_d^\star) & \tau + \delta_d \leq 1, \\ +\infty & \tau + \delta_d > 1. \end{cases}
\end{aligned} \tag{3}$$

For the identification of the quantile functions of the potential outcomes, we impose the following condition.

Assumption 3.2. *For each $d \in \{0,1\}$, the conditional cdfs at the boundary point $r_d^\star$, $F_{Y_d|R}(\cdot \mid r_d^\star)$, are continuous and strictly increasing.*

Assumption 3.2 requires that the potential outcomes be continuous random variables. We now establish the identification of the CQTE.

Theorem 3.5. *Let $r \in Supp(R)$. Suppose tat Assumptions 3.1 and 3.2 hold. For a given $\tau \in (0,1)$,*

the $\text{CQTE}(\tau \mid r)$ *lies in the set*

$$\begin{aligned}&\left[\underline{\text{CQTE}}^{k_c}(\tau \mid r), \overline{\text{CQTE}}^{k_c}(\tau \mid r)\right] \\ \equiv &\begin{cases} \left[Q_{Y|R,D=1}(\tau \mid r) - \overline{Q}^{k_{0,c}}_{Y_0|R}(\tau \mid r), Q_{Y|R,D=1}(\tau \mid r) - \underline{Q}^{k_{0,c}}_{Y_0|R}(\tau \mid r)\right] & \text{for } r \in \mathcal{R}_1, \\ \left[\underline{Q}^{k_{1,c}}_{Y_1|R,D=1}(\tau \mid r) - Q_{Y_0|R}(\tau \mid r), \overline{Q}^{k_{1,c}}_{Y_1|R}(\tau \mid r) - Q_{Y|R,D=0}(\tau \mid r)\right] & \text{for } r \in \mathcal{R}_0, \end{cases}\end{aligned} \tag{4}$$

where $k_c = (k_{0,c}, k_{1,c})$.

Moreover, the interior of this set is sharp.

The main idea of Theorem 3.5 is to invert the bounds on the counterfactual distribution, which is not identified over a certain region of the support of R. Since any quantile is a D_1-parameter in the sense of Stoye (2010) (or D-parameter in the sense of Manski (1997)), the lower (respectively, upper) bound on the conditional quantile is obtained by inverting the upper (respectively, lower) bound on the conditional distribution of the corresponding potential outcome.

The bounds on the CQTE yield the sharp bounds on the CATE, and this is established in the following corollary:

Corollary 3.6. *Let* $r \in Supp(R)$. *Suppose tat Assumptions 3.1 and 3.2 hold. Then, the* $CATE(r)$ *lies in the set*

$$\left[\int_0^1 \underline{\text{CQTE}}^{k_c}(\tau \mid r)d\tau, \int_0^1 \overline{\text{CQTE}}^{k_c}(\tau \mid r)d\tau\right].$$

Moreover, this identified set is sharp.

The unconditional QTEs and ATE can be identified from the result in Corollary 3.4. Specifically, one can derive the bounds on the unconditional QTE, as shown in Theorem 3.5 by inverting the bounds on the unconditional cdf. Once we identify the unconditional QTE for each quantile level $\tau \in (0, 1)$, the unconditional ATE is identified by taking average of the bounds over $\tau \in [0, 1]$.

3.3 Tightening Bounds: Stochastically Monotone Running Variable

It is well recognized that, in RD designs, the conditional expectation functions of potential outcomes on the running variable often exhibit monotonicity. Moreover, a growing body of empirical evidence provides support for the plausibility of such monotonicity in many applied settings (e.g., Lalive (2007); Lee (2008); Duflo et al. (2011); Abdulkadiroğlu et al. (2014); Kaniel and Parham (2017)).[8] In

[8]Motivated by such empirical evidence, several studies have proposed estimation and inference procedures that explicitly incorporate monotonicity restrictions on the functions of interest in RD designs (e.g., Kwon and Kwon (2020); Babii and Kumar (2023)).

addition, it is well established in the literature that monotonicity assumptions can have substantial identifying power (cf. Manski (1997); Manski and Pepper (2000)).

We propose an assumption so called the *stochastically monotone running variable (SMRV)* to tighten the bounds on the counterfactual distribution functions. The SMRV assumption is motivated by the stocahstically monotone instrumental variable assumption in Mourifie et al. (2020) and it places a distributional restriction among the conditional distributions of the potential outcomes on the running variable.

Assumption 3.3 (Stochastically Monotone Running Variable (SMRV))**.** *For any $r, \tilde{r} \in Supp(R)$ such that $r \geq \tilde{r}$, we have, for all $y \in \mathbb{R}$,*

$$F_{Y_d|R}(y|r) \leq F_{Y_d|R}(y|\tilde{r})$$

for $d \in \{0,1\}$.

Assumption 3.3 imposes a first-order stochastic dominance restriction on the conditional distributions of the potential outcomes conditional on the running variable, with the distributions ordered according to the value of the running variable. Stochastic dominance assumptions are widely imposed in the literature and have proven useful for tightening identified bounds (e.g., Blundell et al. (2007); Okumura and Usui (2014); Mourifie et al. (2020); Lee (2024)). Note that the direction of monotonicity may be reversed depending on the empirical context. Moreover, such assumptions are subject to empirical testing.[9] We note that under Assumptions 3.1 and 3.3, bounds on the conditional distribution functions can be derived using arguments analogous to those in Theorem 3.3.

Theorem 3.7. *Suppose that Assumptions 3.1 and 3.3 hold. For each $d \in \{0,1\}$, let $r_d \in \mathcal{R}_d$ be given. Then, we have*

$$F_{Y_1|R}(y|r_0) \in \left[LB_1^{SMRV}(k_{1,c}, r_0), UB_1^{SMRV}(k_{1,c}, r_0)\right],$$
$$F_{Y_0|R}(y|r_1) \in \left[LB_0^{SMRV}(k_{1,c}, r_1), UB_0^{SMRV}(k_{1,c}, r_1)\right],$$

[9] Shen and Zhang (2016) and Qu and Yoon (2019) propose inference methods for distributional treatment effects and, as part of their analysis, consider tests for stochastic dominance in RD designs.

where

$$
\begin{aligned}
LB_1^{SMRV}(k_{1,c}, r_0) &\equiv F_{Y_1|R}(y|r_1^\star), \\
UB_1^{SMRV}(k_{1,c}, r_0) &\equiv \min\left\{F_{Y_1|R}(y|r_1^\star) + k_{1,c} \cdot |r_0 - r_1^\star|, 1\right\}, \\
LB_0^{SMRV}(k_{0,c}, r_1) &\equiv \max\left\{F_{Y_0|R}(y|r_0^\star) - k_{0,c} \cdot |r_1 - r_0^\star|, 0\right\}, \\
UB_0^{SMRV}(k_{0,c}, r_1) &\equiv F_{Y_0|R}(y|r_0^\star),
\end{aligned}
$$

for all $y \in \mathbb{R}$.

It is straightforward to see that the bounds under Assumptions 3.1 and 3.3 are tighter than those in Theorem 3.3 in the sense that $UB_1^{SMRV}(k_{1,c}, r_0) \leq \overline{F}_{Y_1|R}^{k_{1,c}}(y \mid r_0)$ and $LB_0^{SMRV}(k_{0,c}, r_1) \geq \underline{F}_{Y_0|R}^{k_{0,c}}(y \mid r_1)$, implying that the SMRV assumption has substantial identifying power.

Although we focus on identifying the distribution functions of the potential outcomes, our approach can also be applied to identifying their conditional mean functions, in a similar spirit to Kim et al. (2018). A mean version of Assumption 3.3 can therefore be proposed. We provide these results in Appendix D.

4 Estimation and Inference

We now consider estimation and inference for the identified sets of causal parameters, as in Masten et al. (2024), because we are interested in sensitivity analysis with respect to $k_{d,c}$. The main objects of interest are the bounds on the CQTE, CATE, and ATE.[10] We introduce additional notation. For a set A, we denote the space of bounded real-valued functions on A by $l^\infty(A)$. For a sequence of random elements Z_n and a random element Z, we denote the weak convergence of Z_n to Z by $Z_n \Rightarrow Z$.

Let $\{(Y_i, D_i, R_i) : i = 1, 2, ..., n\}$ be a random sample from the population (Y, D, R). For each $d \in \{0, 1\}$, define $\mathcal{J}_d \equiv \{i : R_i \in \mathcal{R}_d\}$ and denote the cardinality of $\mathcal{J}_d$ by n_d. For simplicity, we assume that $Supp(R) = \mathcal{R} = \mathcal{R}_1 \cup \mathcal{R}_0$. The main ingredients for estimating the bounds are the conditional quantile functions $Q_{Y_d|R,D}(\cdot|r, d)$. We impose the following semiparametric specification for these conditional quantile functions.[11]

[10] We also discuss how to estimate and conduct statistical inference on the bounds on the conditional and unconditional distribution functions of the potential outcomes in the Appendix.

[11] Estimation can also be implemented using local nonparametric methods, such as local polynomial estimation. When $r_1^\star = r_0^\star = c$, local polynomial methods such as those developed by Calonico et al. (2014) can be used to estimate the bounds on the conditional expectation functions that are provided in Appendix (see section D). To estimate the bounds on the CQTE in Theorem 3.5, we can employ the approaches proposed by Frandsen et al. (2012) or Shen and Zhang (2016). Nevertheless, we use semiparametric quantile regression estimators for several reasons. First, these

Assumption 4.1. *For each $d \in \{0,1\}$, $\tau \in (0,1)$, and $r \in \mathcal{R}_d$,*

$$Q_{Y_d|R,D}(\tau|r,d) = \mathcal{P}(r)'\beta_d(\tau),$$

where $\mathcal{P}(r)$ is a vector of transformations of r such as polynomials or B-splines and $\beta_d(\tau) \in \mathcal{B} \subseteq \mathbb{R}^{d_\beta}$ with $d_\beta \equiv \dim(\mathcal{P}(r))$.

It is important to note that the parametric specification in Assumption 4.1 is imposed only over $\mathcal{R}_d$; no parametric specification is imposed over $\mathcal{R}_{1-d}$. One can allow $\mathcal{P}(\cdot)$ to differ across $d \in \{0,1\}$. We assume that it is the same across the two groups to simplify the notation. Then, $\beta_d(\tau)$ can be estimated by $\hat{\beta}_d(\tau)$, defined as

$$\hat{\beta}_d(\tau) \equiv \arg\min_{b \in \mathbb{R}^{d_\beta}} \frac{1}{n_d} \sum_{i=1}^{n} \mathbf{1}(D_i = d) \rho_\tau \left(Y_i - \mathcal{P}(R_i)'b\right),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}(u < 0))$. Let $J_d(\tau) \equiv E\left[f_{Y|R,D}\left(\mathcal{P}(R)'\beta_d(\tau)|R,d\right)\mathcal{P}(R)\mathcal{P}(R)'|D=d\right]$ and $\Omega_d \equiv E\left[\mathcal{P}(R)\mathcal{P}(R)'|D=d\right]$ for each $d \in \{0,1\}$, where $f_{Y|R,D}(y|r,d)$ is the conditional density function of Y given $R=r$ and $D=d$. We also let $\mathcal{T}_\beta \subseteq [\epsilon_\beta, 1-\epsilon_\beta]$ be a compact subset of $(0,1)$ for some $\epsilon_\beta \in (0,1/2)$.

Assumption 4.2. *The data $\{(Y_i, D_i, R_i) : i = 1,2,...,n\}$ are i.i.d. and for each $d \in \{0,1\}$, $\frac{n_d}{n} \to p_d \in [\epsilon_p, 1-\epsilon_p]$ for some small $\epsilon_p \in (0,1/2)$.*

Assumption 4.3. *For each $d \in \{0,1\}$, $\mathcal{R}_d$ is a compact subset of $\mathbb{R}$ and $\mathcal{B}$ is a compact subset of $\mathbb{R}^{d_\beta}$.*

Assumption 4.4. *For each $d \in \{0,1\}$, the conditional density $f_{Y|R,D}(y|r,d)$ exists and is bounded and uniformly continuous in y, uniformly over $r \in \mathcal{R}$.*

Assumption 4.5. *For each $d \in \{0,1\}$, $J_d(\tau)$ is positive definite for all $\tau \in \mathcal{T}_\beta$.*

Assumption 4.6. *$\mathcal{P}(\cdot)$ is uniformly continuous and uniformly bounded on $\mathcal{R}_d$ for each $d \in \{0,1\}$ and $E\left[||\mathcal{P}(R)||^{2+\epsilon}\right] < \infty$ for some $\epsilon > 0$.*

Assumption 4.7. *The support of Y is the known compact interval $[\underline{y}, \overline{y}]$.*

estimators have a $\sqrt{n}$ convergence rate under standard conditions, whereas fully nonparametric estimators generally converge more slowly. Second, constructing the CATE and ATE bounds requires evaluating conditional quantiles over many values of τ and r. The semiparametric specification can therefore reduce the computational burden and facilitate the establishment of the asymptotic theory. Moreover, as pointed out in Chernozhukov et al. (2013), a sufficiently rich set of transformations $\mathcal{P}(r)$ can flexibly approximate the conditional quantile function.

Assumptions 4.2–4.6 are standard in the literature on quantile regression (e.g., Angrist et al. (2006)). Assumption 4.3 implies that $r_d^\star \in \mathcal{R}_d$. We also note that Assumption 4.5 is satisfied if $f_{Y|R,D}\left(Q_{Y_d|R,D}(\tau|r,d)|r,d\right)$ is uniformly bounded away from zero over $r \in \mathcal{R}_d$ and $\tau \in \mathcal{T}_\beta$, and $E\left[\mathcal{P}(R)\mathcal{P}(R)'|D=d\right]$ is of full rank. Assumption 4.7 is imposed to ensure that the estimated bounds are finite.

Define $\delta_d(r;k_{d,c}) \equiv k_{d,c}|r - r_d^\star|$ and let

$$s_{d,L}(\tau,r;k_{d,c}) \equiv \max\left(\tau - \delta_d(r;k_{d,c}), \epsilon_s\right),$$
$$s_{d,U}(\tau,r;k_{d,c}) \equiv \min\left(\tau + \delta_d(r;k_{d,c}), 1-\epsilon_s\right),$$

where $\epsilon_s \in (0,1/2)$ is a small fixed trimming constant used only for estimation and inference. The constant ϵ_s need not be the same as ϵ_β. These transformed quantile indices contain no estimated quantities. For each $d \in \{0,1\}$, let $\mathcal{T}_{C,d}$ be a compact subset of $\mathcal{T}_\beta$ such that

$$\tau - \delta_{1-d}(r;k_{1-d,c}), \quad \tau + \delta_{1-d}(r;k_{1-d,c}) \in \mathcal{T}_\beta \cap [\epsilon_s, 1-\epsilon_s]$$

uniformly over $(\tau,r) \in \mathcal{T}_{C,d} \times \mathcal{R}_d$.[12]

Recall that $\underline{Q}_d^{k_{d,c}}(\tau \mid r) = \mathcal{P}(r_d^\star)'\beta_d\left(s_{d,L}(\tau,r;k_{d,c})\right)$ and $\overline{Q}_d^{k_{d,c}}(\tau \mid r) = \mathcal{P}(r_d^\star)'\beta_d\left(s_{d,U}(\tau,r;k_{d,c})\right)$ for $r \in \mathcal{R}_{1-d}$. We can construct their estimators by replacing $\beta_d(\cdot)$ with $\hat{\beta}_d(\cdot)$. The bounds on the CQTE contain the conditional quantile function of Y given $D=d$ and $R=r \in \mathcal{R}_d$. This component can be estimated using the quantile regression estimator of $\beta_d(\cdot)$. The following theorem establishes the limiting distributions of the estimated bounds on the CQTE.

Theorem 4.1. *Suppose that Assumptions 3.1, 3.2, and 4.1–4.7 hold. For each $d \in \{0,1\}$,*

$$\sqrt{n}\begin{pmatrix} \widehat{\underline{CQTE}}_d(\tau,r) - \underline{CQTE}_d(\tau,r) \\ \widehat{\overline{CQTE}}_d(\tau,r) - \overline{CQTE}_d(\tau,r) \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbb{Z}_{CQ,L,d}(\tau,r) \\ \mathbb{Z}_{CQ,U,d}(\tau,r) \end{pmatrix}$$

[12] The subscript d in $\mathcal{T}_{C,d}$ indexes the region to which r belongs. By construction, the truncation in $s_{1-d,L}$ and $s_{1-d,U}$ does not bind on $\mathcal{T}_{C,d} \times \mathcal{R}_d$. Thus, both τ and the corresponding untruncated transformed quantile indices for the counterfactual potential outcome lie in $\mathcal{T}_\beta$, the range over which weak convergence of the primitive quantile regression estimators is established.

in $l^\infty(\mathcal{T}_{C,d} \times \mathcal{R}_d)^2$*, where for* $r \in \mathcal{R}_1$ *and* $\tau \in \mathcal{T}_{C,1}$*,*

$$\mathbb{Z}_{CQ,L,1}(\tau, r) \equiv \mathcal{P}(r)'\mathbb{Z}_1(\tau) - \mathcal{P}(r_0^*)'\mathbb{Z}_0\left(s_{0,U}(\tau, r; k_{0,c})\right),$$
$$\mathbb{Z}_{CQ,U,1}(\tau, r) \equiv \mathcal{P}(r)'\mathbb{Z}_1(\tau) - \mathcal{P}(r_0^*)'\mathbb{Z}_0\left(s_{0,L}(\tau, r; k_{0,c})\right),$$

and for $r \in \mathcal{R}_0$ *and* $\tau \in \mathcal{T}_{C,0}$*,*

$$\mathbb{Z}_{CQ,L,0}(\tau, r) \equiv \mathcal{P}(r_1^*)'\mathbb{Z}_1\left(s_{1,L}(\tau, r; k_{1,c})\right) - \mathcal{P}(r)'\mathbb{Z}_0(\tau),$$
$$\mathbb{Z}_{CQ,U,0}(\tau, r) \equiv \mathcal{P}(r_1^*)'\mathbb{Z}_1\left(s_{1,U}(\tau, r; k_{1,c})\right) - \mathcal{P}(r)'\mathbb{Z}_0(\tau),$$

Here, $\mathbb{Z}_d(\cdot)$ *is a mean-zero Gaussian process whose covariance kernel is given in the Appendix.*

We now consider estimation and inference for the bounds on the CATE and ATE. To construct the CATE bounds, we extend the CQTE bounds outside $\mathcal{T}_{C,d}$ using the known support endpoints while retaining the observed conditional quantile whenever $\tau \in \mathcal{T}_\beta$. That is, for $r \in \mathcal{R}_1$ and $\tau \in \mathcal{T}_\beta \setminus \mathcal{T}_{C,1}$, define

$$\underline{CQTE}_1(\tau, r) \equiv Q_{Y|R,D=1}(\tau|r) - \overline{y},$$
$$\overline{CQTE}_1(\tau, r) \equiv Q_{Y|R,D=1}(\tau|r) - \underline{y}.$$

Similarly, for $r \in \mathcal{R}_0$ and $\tau \in \mathcal{T}_\beta \setminus \mathcal{T}_{C,0}$, define

$$\underline{CQTE}_0(\tau, r) \equiv \underline{y} - Q_{Y|R,D=0}(\tau|r),$$
$$\overline{CQTE}_0(\tau, r) \equiv \overline{y} - Q_{Y|R,D=0}(\tau|r).$$

For $\tau \in [0,1] \setminus \mathcal{T}_\beta$ and each $d \in \{0,1\}$, define

$$\underline{CQTE}_d(\tau, r) \equiv \underline{y} - \overline{y},$$
$$\overline{CQTE}_d(\tau, r) \equiv \overline{y} - \underline{y}.$$

The corresponding estimators are defined analogously, replacing the observed conditional quantile by its QR estimator for $\tau \in \mathcal{T}_\beta$ and treating the support endpoints as known constants. Under

these extensions, for each $d \in \{0,1\}$ and $r \in \mathcal{R}_d$, define the lower and upper CATE bounds by

$$\underline{CATE}_d(r) \equiv \int_0^1 \underline{CQTE}_d(\tau, r)d\tau,$$
$$\overline{CATE}_d(r) \equiv \int_0^1 \overline{CQTE}_d(\tau, r)d\tau.$$

Their estimators are defined by integrating the corresponding estimated CQTE bounds. In implementation, these integrals may be approximated using a deterministic grid over τ.[13]

Theorem 4.2. *Suppose that Assumptions 3.1, 3.2, and 4.1–4.7 hold. For each $d \in \{0,1\}$,*

$$\sqrt{n}\begin{pmatrix} \widehat{\underline{CATE}}_d(r) - \underline{CATE}_d(r) \\ \widehat{\overline{CATE}}_d(r) - \overline{CATE}_d(r) \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbb{Z}_{C,L,d}(r) \\ \mathbb{Z}_{C,U,d}(r) \end{pmatrix}$$

in $l^\infty(\mathcal{R}_d)^2$, where

$$\mathbb{Z}_{C,L,d}(r) \equiv \int_{\mathcal{T}_{C,d}} \mathbb{Z}_{CQ,L,d}(\tau, r)d\tau + (2d-1)\int_{\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}} \mathcal{P}(r)'\mathbb{Z}_d(\tau)d\tau,$$
$$\mathbb{Z}_{C,U,d}(r) \equiv \int_{\mathcal{T}_{C,d}} \mathbb{Z}_{CQ,U,d}(\tau, r)d\tau + (2d-1)\int_{\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}} \mathcal{P}(r)'\mathbb{Z}_d(\tau)d\tau.$$

Recall that the lower and upper ATE bounds are $\underline{ATE} = \sum_{d=0}^1 \int_{\mathcal{R}_d} \underline{CATE}_d(r)dF_R(r)$ and $\overline{ATE} = \sum_{d=0}^1 \int_{\mathcal{R}_d} \overline{CATE}_d(r)dF_R(r)$. Let $\widehat{F}_R$ denote the empirical distribution of R. We estimate the ATE bounds by

$$\begin{aligned}
\widehat{\underline{ATE}} &\equiv \sum_{d=0}^1 \int_{\mathcal{R}_d} \widehat{\underline{CATE}}_d(r)d\widehat{F}_R(r) \\
&= \frac{1}{n}\sum_{i=1}^n \sum_{d=0}^1 \mathbf{1}\{R_i \in \mathcal{R}_d\}\widehat{\underline{CATE}}_d(R_i), \\
\widehat{\overline{ATE}} &\equiv \sum_{d=0}^1 \int_{\mathcal{R}_d} \widehat{\overline{CATE}}_d(r)d\widehat{F}_R(r) \\
&= \frac{1}{n}\sum_{i=1}^n \sum_{d=0}^1 \mathbf{1}\{R_i \in \mathcal{R}_d\}\widehat{\overline{CATE}}_d(R_i).
\end{aligned}$$

[13] We assume that the numerical integration error is $o(n^{-1/2})$ uniformly in r.

Theorem 4.3. *Suppose that Assumptions 3.1, 3.2, and 4.1–4.7 hold. Then,*

$$\sqrt{n}\left(\widehat{\underline{ATE}} - \underline{ATE}\right) \Rightarrow \sum_{d=0}^{1} \int_{\mathcal{R}_d} \mathbb{Z}_{C,L,d}(r) dF_R(r) + \mathbb{G}_{R,L},$$
$$\sqrt{n}\left(\widehat{\overline{ATE}} - \overline{ATE}\right) \Rightarrow \sum_{d=0}^{1} \int_{\mathcal{R}_d} \mathbb{Z}_{C,U,d}(r) dF_R(r) + \mathbb{G}_{R,U},$$

where $(\mathbb{G}_{R,L}, \mathbb{G}_{R,U})$ *is a mean-zero Gaussian vector defined in the Appendix and is generally correlated with the CATE limit processes.*

We conclude this section with a few remarks on the additional issues involved in inference for the bounds on the unconditional QTE. Note that the unconditional QTE bounds can be obtained by inverting the bounds on the unconditional distribution functions, and the latter involve max and min operators, which are Hadamard directionally differentiable (cf. Fang and Santos (2019)). The lack of differentiability of the bounding functions with respect to the observed distribution functions precludes the use of the standard delta method for the inverse map (cf. Lemma 3.9.20 in Van Der Vaart and Wellner (1996)). Establishing the asymptotic distribution of the estimated bounds on the unconditional QTE is beyond the scope of this paper, and we leave this important issue as future research.

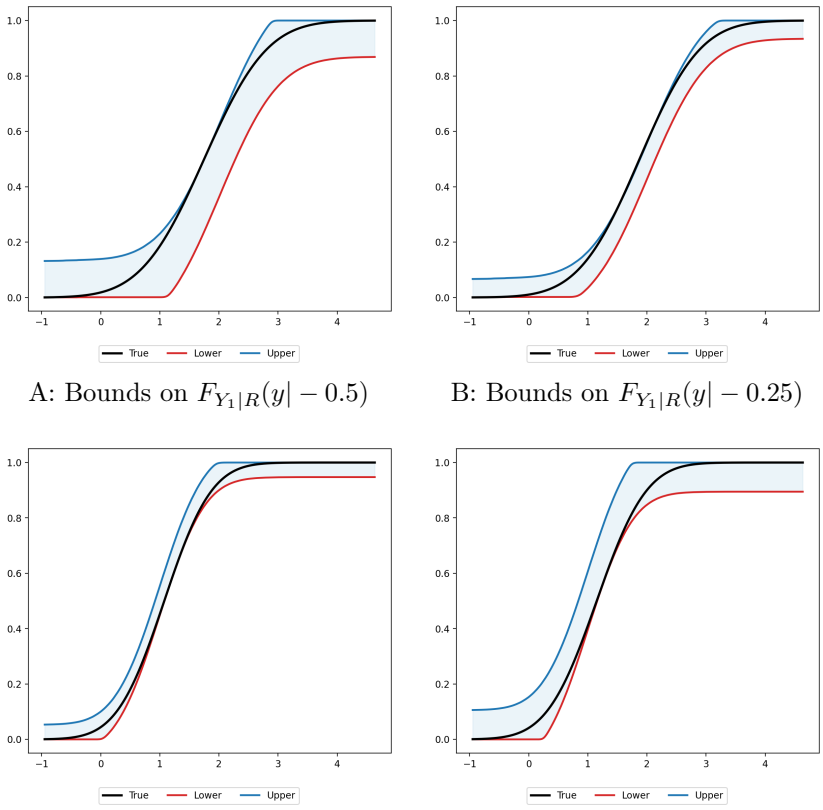
5 Simulation

We conduct a Monte Carlo simulation to investigate the finite sample performance of the identification results. In the Monte Carlo simulation, the potential outcomes and treatment variable are generated as follows:

$$Y_1 = g_1(R) + s_1(R) \cdot \epsilon_1,$$
$$Y_0 = g_0(R) + s_0(R) \cdot \epsilon_0,$$
$$D = \mathbf{1}\left(R \geq 0\right),$$

where $(\epsilon_1, \epsilon_0)' | R \sim BVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.2 \\ 0.2 & 1 \end{pmatrix}\right)$, $R \sim Uniform(-1, 1)$, $g_1(r) = 2 + 0.5 \cdot r$, $g_0(r) = 1 + 0.3r$, $s_1(r) = 0.8 + (0.4 \cdot r)^2$ and $s_0(r) = 0.6 + (0.1 + 0.3 \cdot r)^2$. This data generating profcess (DGP) yields a location-scale shift quantile regression model, which generates different shapes of the conditional distribution (cf. Koenker and Xiao (2002)). We generate 5,000 observations for

Figure 5.1: Bounds on Conditional CDFs: Location-Scale Shift Model



A: Bounds on $F_{Y_1|R}(y|-0.5)$ B: Bounds on $F_{Y_1|R}(y|-0.25)$

C: Bounds on $F_{Y_0|R}(y|0.25)$ D: Bounds on $F_{Y_0|R}(y|0.5)$

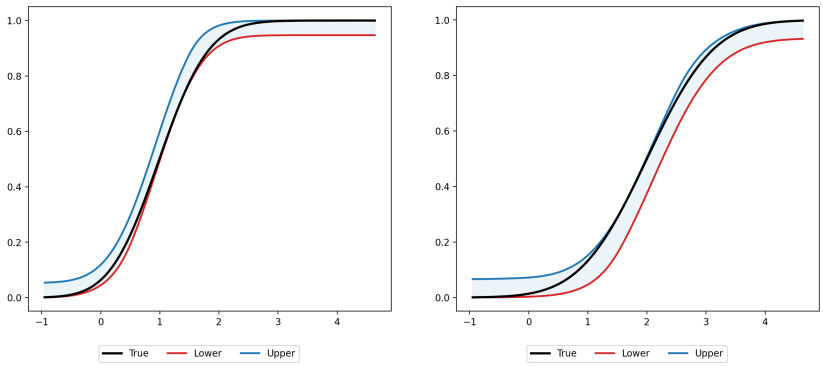
Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

each simulation, and the number of replications is set to be 1,000.

To estimate the conditional distributions, we employ the DR with $\mathcal{P}(x) = x$. The grid size over $\mathcal{Y}_d$ is set to be 0.01, and we set $\mathcal{Y}_1 = \mathcal{Y}_d$ for simplicity. To construct the bounds on the conditional distributions, it is necessary to choose the nuisance parameters $k_{d,c}$ for $d \in \{0, 1\}$.Since the derivative of the conditional distribution over the region where it is identified is given by $\lambda\left(r \cdot \beta_d(y)\right)\beta_d(y)$, we set $k_{d,c}$ equal to the maximum absolute value of this derivative over the identified region.

Figure 5.1 reports the simulation results on the bounds on the conditional distributions of the potential outcomes. We consider four different conditioning values, $r \in \{\pm 0.25, \pm 0.5\}$. The blue lines are pointwise average of the upper bounds on $F_{Y_1|R}(\cdot|r)$ for $r \in \{-0.5, -0.25\}$ over the simulation repetitions, and the red lines depict pointwise average of the lower bounds on $F_{Y_0|R}(\cdot|r)$ for $r \in \{0.25, 0.5\}$. We can find that the bounds contain the true distribution function in all cases. In addition, the (average) upper bounds on $F_{Y_1|R}$ are very close to the true conditional distribution over some region of $\mathcal{Y}_1$. A similar pattern can be found for the lower bounds on $F_{Y_0|R}$.

Figure 5.2: Bounds on Unconditional CDFs: Location-Scale Shift Model



A: Bounds on $F_{Y_0}(y)$ B: Bounds on $F_{Y_1}(y)$

Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

Figure 5.2 show the estimated bounds on the unconditional distribution functions F_{Y_1} and F_{Y_0}. As explained earlier, these bounds are obtained by taking average of the conditional distribution functions over $Supp(R)$. We can see that the bounds on the unconditional distributions are much tighter than those on the conditional distributions, and this is consistent with the argument in Firpo and Ridder (2019).[14]

We report the simulation results for causal parameters such as the CQTE, CATE, QTE, and ATE in Table 5.1. All estimated bounds contain the true value of the corresponding causal parameter. We also observe that the upper bounds tend to be farther from the true parameter values than the lower bounds.

We investigate the identifying power of the SMRV assumption. The DGP is modified slightly so that it satisfies the SMRV assumption. Specifically, we set $s_1(r) = s_0(r) = 1$, which yields a location shift model that satisfies the SMRV assumption. Tables 5.2 and 5.3 report the simulation results for the CQTE and QTE. The baseline bounds are estimated without imposing the SMRV assumption, whereas the SMRV bounds are estimated under the SMRV assumption. It is evident that the bounds under the SMRV assumption contain the corresponding causal parameter and are much tighter than the baseline bounds. For all causal parameters, the upper bounds are substantially lower, resulting in much narrower bounds. The lower bounds under the SMRV assumption are the same to the baseline lower bounds, which reflects the fact that the SMRV assumption imposes

[14] Firpo and Ridder (2019) show that when the potential outcomes are correlated with an observed variable, the bounds on the (unconditional) distribution of treatment effects can be tightened by averaging over the values of that covariate. Although the role of the running variable in sharp RD designs differs from that of observed covariates in randomized experiments, the tightening of the bounds on the unconditional distributions of the potential outcomes in our framework may reflect a similar idea.

Table 5.1: Bounds on Causal Parameters: Location-Scale Shift Model

Type	τ	r	True	Estimated bounds
CQTE				
CQTE	0.1	-0.5	0.596	$[-1.002,\ 1.358]$
CQTE	0.1	0.5	0.873	$[0.648,\ 2.080]$
CQTE	0.3	-0.5	0.775	$[0.723,\ 1.342]$
CQTE	0.3	0.5	1.007	$[0.943,\ 1.314]$
CQTE	0.5	-0.5	0.900	$[0.896,\ 1.425]$
CQTE	0.5	0.5	1.100	$[1.084,\ 1.405]$
CQTE	0.7	-0.5	1.025	$[0.980,\ 1.597]$
CQTE	0.7	0.5	1.193	$[1.171,\ 1.544]$
CQTE	0.9	-0.5	1.204	$[0.963,\ 3.155]$
CQTE	0.9	0.5	1.327	$[-1.458,\ 1.836]$
QTE				
QTE	0.1	–	0.727	$[0.353,\ 1.396]$
QTE	0.3	–	0.888	$[0.829,\ 1.328]$
QTE	0.5	–	1.000	$[0.979,\ 1.416]$
QTE	0.7	–	1.112	$[1.068,\ 1.573]$
QTE	0.9	–	1.274	$[1.091,\ 2.124]$
CATE				
CATE	–	-0.5	0.900	$[0.661,\ 1.644]$
CATE	–	0.5	1.100	$[0.844,\ 1.521]$
ATE				
ATE	–	–	1.000	$[0.844,\ 1.557]$

Note: The nuisance parameters are set to $k_0 = 0.211$ and $k_1 = 0.262$.

Table 5.2: Identifying Power of the SMRV Assumption: CQTE

	CQTE at $r = -0.5$			CQTE at $r = 0.5$		
τ	True	Baseline	SMRV	True	Baseline	SMRV
0.1	0.900	[−0.753, 1.564]	[−0.753, 1.109]	1.100	[0.966, 1.717]	[0.966, 1.252]
0.3	0.900	[0.813, 1.406]	[0.813, 1.135]	1.100	[1.083, 1.432]	[1.083, 1.249]
0.5	0.900	[0.885, 1.396]	[0.885, 1.142]	1.100	[1.096, 1.398]	[1.096, 1.248]
0.7	0.900	[0.875, 1.465]	[0.875, 1.147]	1.100	[1.062, 1.411]	[1.062, 1.244]
0.9	0.900	[0.711, 2.977]	[0.711, 1.154]	1.100	[0.777, 1.529]	[0.777, 1.244]

Note: The nuisance parameters are set to $k_0 = 0.120$ and $k_1 = 0.199$. The true DGP is a location shift model, where $s_1(r) = s_0(r) = 1$.

monotonicity on the conditional distributions only in one direction. Nevertheless, the simulation results confirm that the SMRV assumption has substantial identifying power.

Table 5.3: Identifying Power of the SMRV Assumption: QTE

τ	True	Baseline	SMRV
0.1	1.000	[0.636, 1.613]	[0.636, 1.169]
0.3	1.000	[0.948, 1.414]	[0.948, 1.185]
0.5	1.000	[0.991, 1.398]	[0.991, 1.193]
0.7	1.000	[0.973, 1.440]	[0.973, 1.203]
0.9	1.000	[0.812, 1.740]	[0.812, 1.214]

Note: The nuisance parameters are set to $k_0 = 0.120$ and $k_1 = 0.199$. The true DGP is a location shift model, where $s_1(r) = s_0(r) = 1$.

6 Empirical Application

We provide an empirical illustration that considers the effect of better schools on student's academic performance to demonstrate how the proposed methods can be applied in practice. When considering the effect of attending a better school on student performance, it is well documented in the literature that the effect for students who are marginally admitted to the school may differ from the effect for those who qualify more easily (e.g., Abdulkadiroğlu et al. (2014); Dobbie and Fryer Jr (2014); Angrist and Rokkanen (2015)). For this reason, it is worthwhile to investigate the causal effects at values of the running variable away from the cutoff, and we apply the proposed methods to shed light on potential heterogeneity in the causal effect across the running variable.

We revisit the empirical study of Pop-Eleches and Urquiola (2013), who examine the effect of attending a better school on student performance in the Romanian secondary school system. In Romania, admission from middle school to high school is determined by students' transition scores,

which are based on their performance on a national eighth-grade exam, together with their ranked lists of preferred high schools. As described in Pop-Eleches and Urquiola (2013), this assignment process generates a cutoff score for each school, which makes it possible to use an RD design to identify the causal effect of access to higher-achieving schools. Among the outcomes they consider is students' score on the Baccalaureate exam. This outcome is particularly important because students must pass the exam to be eligible to apply to university, and the exam score is also used as a key factor in the admissions process.

We use the dataset from Shen and Zhang (2016), which is identical to that used in Pop-Eleches and Urquiola (2013), except that Shen and Zhang (2016) restrict the sample to two-school towns to mitigate estimation bias caused by multiple cutoff values. The outcome variable Y is the grade of the Baccalaureate exam, and the running variable R is the transition score centered at the cutoff value. Table 6.1 reports the summary statistics of the dataset, and Figure A.1 depict the kernel density estimators for R and Y. While the cutoff value is set to be zero, more than half of students in the sample (approximately 72%) are above the cutoff value. The distribution of the transition score is is not symmetric around the cutoff value, and it is not uniformly distributed.

Table 6.1: Summary Statistics

Variable	n	Mean	Std.dev	Min	Q25	Q50	Q75	Max
Baccalaureate grade (Y)	14,765	-0.06	0.80	-2.95	-0.62	0.01	0.55	2.00
Transition score (R)	14,765	0.48	0.75	-1.93	-0.09	0.49	1.07	1.93
Treatment (D)	14,765	0.72	0.45	0.00	0.00	1.00	1.00	1.00

Note: The running variable is the transition score centered at the admission cutoff, so that the cutoff value is zero; that is, $D = \mathbf{1}(R \geq 0)$. The Baccalaureate exam grade is the outcome variable.

The main ingredients to estimate the bounds are the conditional distributions of the potential outcomes on the running variable, and the distribution functions are evaluated at the boundary points of $\mathcal{R}_1$ and $\mathcal{R}_0$. As in the simulation study, we consider $\mathcal{R}_1 = [c, r_{max}]$ and $\mathcal{R}_0 = [r_{min}, c)$, and thus, the boundary points are the cutoff level c. We normalize the cutoff value to zero. Since our extrapolation is based on the boundary points of $\mathcal{R}_1$ and $\mathcal{R}_0$, which are both equal to c, we first compare the DR estimators evaluated at $R = c$ with the corresponding nonparametric estimators proposed by Calonico et al. (2014) As shown in Figure A.2 in Appendix, the difference between these two estimators is negligible. This comparison shows that the empirical results are not sensitive to the choice of estimation methods.

We focus on the identification and estimation of causal parameters, including the CQTE, QTE, CATE, and ATE. Table 6.2 reports the estimated bounds on the CQTE for different quantile levels

and values of the running variable. To facilitate comparison, we also report the RD estimates of the QTE at the cutoff, that is, the CQTE at $R = 0$. In addition, we compare the estimated bounds obtained with and without the SMRV assumption. For $R = r$ such that $r < 0$, the estimated bounds contain the RD estimates in all cases. Moreover, the bounds include zero, so we cannot determine the sign of the CQTE conditional on $R = r$ when $r = 0$. In contrast, for $R = r$ such that $r > 0$, some estimated bounds on the CQTE conditional on $R = r$ do not contain the RD estimate of the QTE at the cutoff. For example, the estimated lower bound on the 0.1th CQTE given $R = 0.5$ is 0.227 both with and without the SMRV assumption, whereas the RD estimate at the cutoff is 0.182. Similar results can be found for different quantile levels. The estimated lower bounds on the τ-th CQTE given $R = 0.5$ are greater than the RD estimate at the cutoff except for $\tau \in \{0.8, 0.9\}$. Setting aside sampling error, this result suggests that the 0.1th CQTE given $R = 0.5$ differs from the QTE at the cutoff for the same quantile level. Similar patterns are observed for other values of r. In particular, the lower bounds tend to increase as the conditioning value r increases.

Figure 6.1 plots the estimated bounds, with and without the SMRV assumption, for each fixed quantile level. The figure clearly shows that the SMRV assumption does not improve the lower bounds, but it substantially improves the upper bounds, thereby yielding narrower identified sets. Consistent with the results in Table 6.2, these findings suggest that the effect of attending a better school on performance on the Baccalaureate exam may be heterogeneous across subpopulations defined by the value of the running variable, as some bounds on the CQTE at values of the running variable different from the cutoff do not contain the corresponding RD estimates.

In Figure A.3 in Appendix, we report the estimated bounds on the CQTE for various values of k_0 and k_1 to see the sensitivity of the bounds to these nuisance parameters. Consistent with our theoretical predictions, the bounds become wider as either k_0 or k_1 increases.

We now consider the estimated bounds on the unconditional QTE. Table 6.3 presents the estimated bounds on the unconditional QTE as well as the RD estimate of the QTE at $R = 0$. While the bounds at $\tau \in \{0.1, 0.9\}$ are uninformative in the sense that the sign of QTE is not identified, we can identify the sign of QTE at other quantile levels. An interesting finding is that, for τ ranging from 0.2 to 0.7, the lower bounds are greater than the RD estimates. This implies that the QTE of attending a better school on Baccalaureate grades for *the whole population* may differ from that for *students at the margin* of admission to a better high school.

Using the estimated bounds on the CQTE and QTE, we derive bounds on the CATE and ATE. The RD estimate, which corresponds to the ATE given $R = 0$, is 0.2. The estimated bounds exhibit heterogeneity across values of the running variable. In particular, the sign of the CATE is identified

Table 6.2: Bounds on CQTE

τ	CQTE at $r=0$	CQTE at $r=-1$		CQTE at $r=-0.5$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	$[-0.627,\ 1.575]$	$[-0.627,\ 0.522]$	$[-0.807,\ 0.953]$	$[-0.807,\ 0.342]$
0.2	0.208	$[-0.930,\ 1.474]$	$[-0.930,\ 0.534]$	$[-1.112,\ 0.833]$	$[-1.112,\ 0.352]$
0.3	0.232	$[-1.159,\ 1.451]$	$[-1.159,\ 0.559]$	$[-0.247,\ 0.838]$	$[-0.247,\ 0.401]$
0.4	0.211	$[-1.343,\ 1.510]$	$[-1.343,\ 0.557]$	$[-0.106,\ 0.849]$	$[-0.106,\ 0.393]$
0.5	0.220	$[-0.538,\ 1.710]$	$[-0.538,\ 0.602]$	$[-0.025,\ 0.863]$	$[-0.025,\ 0.417]$
0.6	0.201	$[-0.365,\ 2.499]$	$[-0.365,\ 0.619]$	$[-0.015,\ 0.911]$	$[-0.015,\ 0.439]$
0.7	0.183	$[-0.269,\ 2.315]$	$[-0.269,\ 0.628]$	$[-0.012,\ 1.005]$	$[-0.012,\ 0.423]$
0.8	0.212	$[-0.288,\ 2.103]$	$[-0.288,\ 0.639]$	$[-0.053,\ 1.888]$	$[-0.053,\ 0.425]$
0.9	0.182	$[-0.372,\ 1.810]$	$[-0.372,\ 0.616]$	$[-0.107,\ 1.629]$	$[-0.107,\ 0.435]$
τ	CQTE at $r=0$	CQTE at $r=0.5$		CQTE at $r=1$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	$[0.227,\ 1.551]$	$[0.227,\ 0.582]$	$[0.299,\ 1.917]$	$[0.299,\ 0.948]$
0.2	0.208	$[0.332,\ 1.028]$	$[0.332,\ 0.620]$	$[0.427,\ 2.241]$	$[0.427,\ 0.984]$
0.3	0.232	$[0.364,\ 0.924]$	$[0.364,\ 0.632]$	$[0.424,\ 1.784]$	$[0.424,\ 0.972]$
0.4	0.211	$[0.342,\ 0.882]$	$[0.342,\ 0.614]$	$[0.382,\ 1.521]$	$[0.382,\ 0.934]$
0.5	0.220	$[0.306,\ 0.841]$	$[0.306,\ 0.589]$	$[0.339,\ 1.429]$	$[0.339,\ 0.890]$
0.6	0.201	$[0.268,\ 0.814]$	$[0.268,\ 0.532]$	$[0.226,\ 1.347]$	$[0.226,\ 0.811]$
0.7	0.183	$[0.232,\ 0.753]$	$[0.232,\ 0.489]$	$[-0.058,\ 1.301]$	$[-0.058,\ 0.750]$
0.8	0.212	$[0.039,\ 0.731]$	$[0.039,\ 0.477]$	$[-0.961,\ 1.241]$	$[-0.961,\ 0.716]$
0.9	0.182	$[-0.959,\ 0.782]$	$[-0.959,\ 0.419]$	$[-0.751,\ 1.246]$	$[-0.751,\ 0.627]$
τ	CQTE at $r=0$	CQTE at $r=1.25$		CQTE at $r=1.5$	
		Baseline	SMRV	Baseline	SMRV
0.1	0.182	$[0.337,\ 2.084]$	$[0.337,\ 1.115]$	$[0.381,\ 2.252]$	$[0.381,\ 1.282]$
0.2	0.208	$[0.442,\ 2.394]$	$[0.442,\ 1.137]$	$[0.448,\ 2.544]$	$[0.448,\ 1.286]$
0.3	0.232	$[0.439,\ 2.604]$	$[0.439,\ 1.118]$	$[0.451,\ 2.749]$	$[0.451,\ 1,264]$
0.4	0.211	$[0.406,\ 1.892]$	$[0.406,\ 1.085]$	$[0.413,\ 1.584]$	$[0.413,\ 1.218]$
0.5	0.220	$[0.340,\ 1.707]$	$[0.340,\ 1.022]$	$[0.271,\ 2.042]$	$[0.271,\ 1.163]$
0.6	0.201	$[0.135,\ 1.623]$	$[0.135,\ 0.949]$	$[-0.219,\ 1.906]$	$[-0.219,\ 1.074]$
0.7	0.183	$[-0.998,\ 1.547]$	$[-0.998,\ 0.871]$	$[-0.879,\ 1.801]$	$[-0.879,\ 0.990]$
0.8	0.212	$[-0.851,\ 1.507]$	$[-0.851,\ 0.826]$	$[-0.751,\ 1.739]$	$[-0.751,\ 0.926]$
0.9	0.182	$[-0.647,\ 1.476]$	$[-0.647,\ 0.731]$	$[-0.553,\ 1.736]$	$[-0.553,\ 0.825]$

Note: The cutoff value is zero. "Baseline" and "SMRV" denote the corresponding bounds on the CQTE given $R = r$ without and with the SMRV assumption, respectively. The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

Figure 6.1: Bounds on the CQTE for Different Values of $R = r$

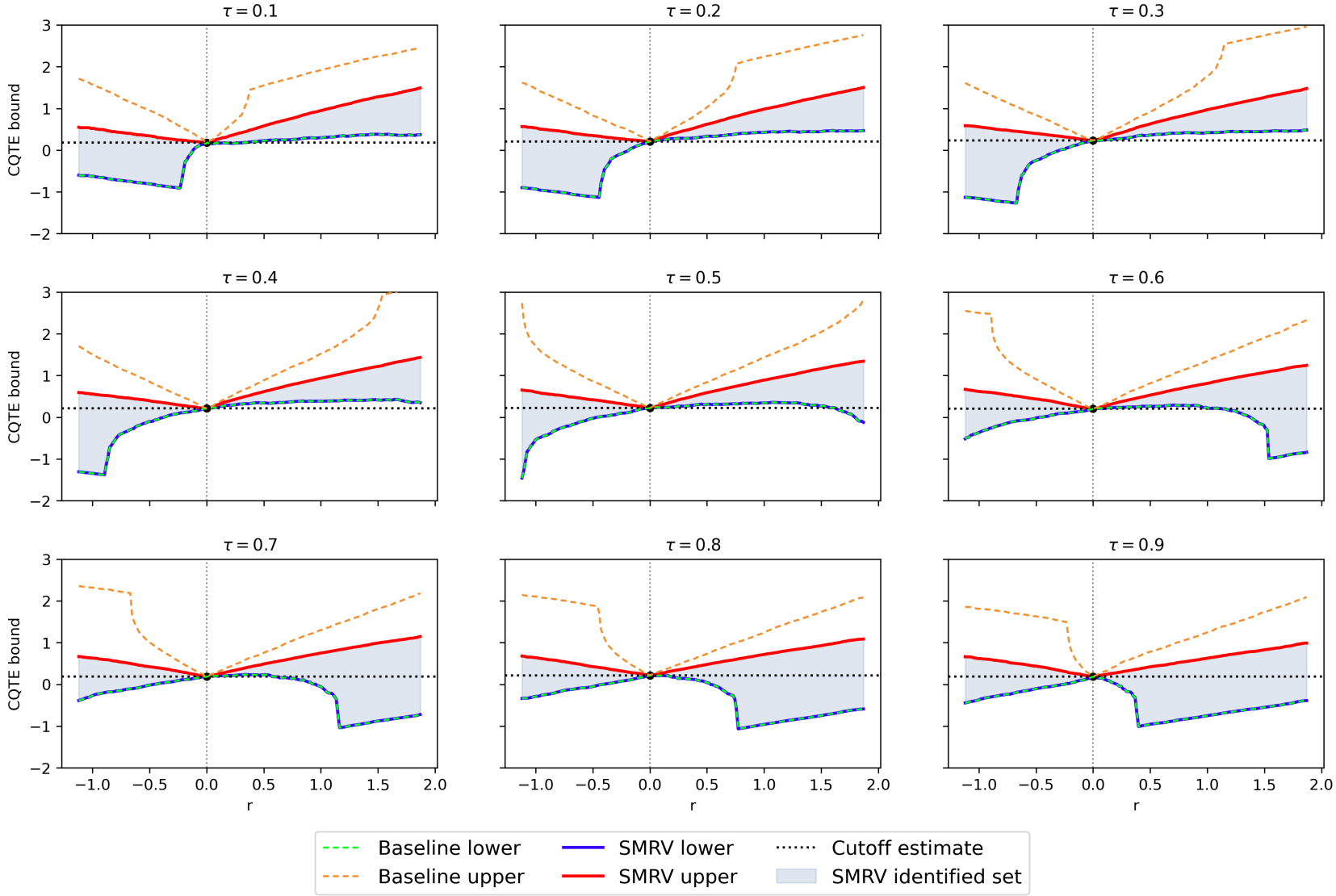


Table 6.3: Bounds on the QTE

τ	RD Estimate	QTE	
		Baseline	SMRV
0.1	0.182	[−0.117, 1.668]	[−0.117, 0.524]
0.2	0.208	[0.119, 1.331]	[0.119, 0.601]
0.3	0.232	[0.195, 1.159]	[0.195, 0.660]
0.4	0.211	[0.244, 1.142]	[0.244, 0.683]
0.5	0.220	[0.257, 1.113]	[0.257, 0.698]
0.6	0.201	[0.254, 1.110]	[0.254, 0.666]
0.7	0.183	[0.241, 1.114]	[0.241, 0.644]
0.8	0.212	[0.014, 1.108]	[0.014, 0.654]
0.9	0.182	[−0.846, 1.186]	[−0.846, 0.596]

Note: The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

for $r \in \{0.5, 1\}$, and the sign of the ATE for the whole population is also identified. Nevertheless, we do not find evidence that the ATE for students at the margin of admission to a better high school differs from that for other subpopulation of students.

Table 6.4: Bounds on the CATE and ATE

	r	Baseline	SMRV
CATE	-1	[−0.668, 1.833]	[−0.668, 0.587]
CATE	-0.5	[−0.270, 1.074]	[−0.270, 0.403]
CATE	0.5	[0.132, 0.919]	[0.132, 0.544]
CATE	1	[0.023, 1.568]	[0.023, 0.843]
CATE	1.25	[−0.028, 1.864]	[−0.028, 0.976]
CATE	1.5	[−0.076, 2.147]	[−0.076, 1.110]
ATE	–	[0.010, 1.230]	[0.010, 0.640]

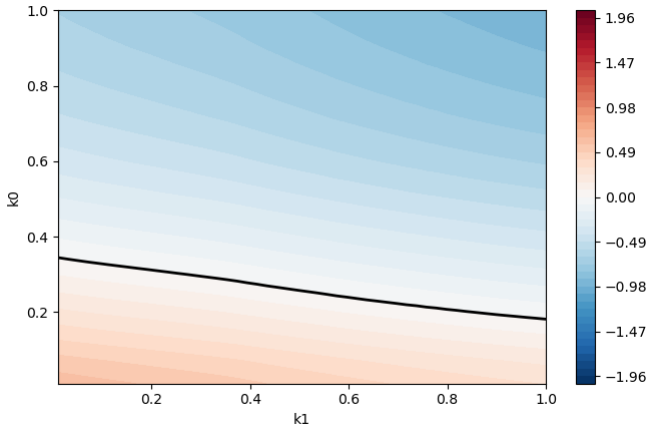
Note: The RD estimate is 0.2. The nuisance parameters are set to $k_0 = 0.26$ and $k_1 = 0.45$.

We then conduct a breakdown frontier analysis for the ATE. The purpose of breakdown frontier analysis is to determine the maximal degree to which the underlying assumptions can be relaxed while still preserving a given conclusion.[15] We seek the combinations of k_0 and k_0 that yield the lower bound on the ATE for the whole population is greater than or equal to zero. In this case, the breakdown frontier can be characterized by the set of values of (k_0, k_1) such that the lower bound on the ATE is greater than or equal to zero.

Figure 6.2 presents the breakdown frontier for the conclusion $ATE \geq 0$. The red region represents the set of (k_0, k_1) values for which the estimated lower bound on the ATE is nonnegative. The blue region represents the set of (k_0, k_1) values for which the estimated lower bound on the

[15] One can refer to Masten and Poirier (2020), who develop an approach to inference for breakdown frontiers for relaxation of the random assignment of treatment assumption and rank invariance, for a detailed discussion on breakdown frontier analysis.

Figure 6.2: Breakdown Frontier for $ATE \geq 0$



Note: The x-axis indexes the values of k_1, and y-axis indexes the values of k_0. The red area is the set of (k_0, k_1) such that the estimated lower bound on ATE is greater than or equal to zero. The blue area is the set of (k_0, k_1) such that the estimated lower bound on ATE is less than zero. The black solid line is the breakdown frontier.

ATE is negative. Accordingly, the black solid line in the figure denotes the breakdown frontier.

To maintain the conclusion that $ATE \geq 0$, the maximum value of k_0 when the potential outcome Y_1 and the running variable R are independent (i.e., $k_1 = 0$), is approximately 0.35. On the other hand, when we do not impose any restriction on the dependence between Y_1 and R (i.e., $k_1 = 1$), the maximal relaxation of the independence between Y_0 and R is about 0.2. The breakdown frontier in Figure 6.2 clearly illustrates the trade-off between the two assumptions on the dependence between the potential outcomes and the running variable required to maintain a given conclusion.

7 Conclusions

In this paper, we propose a framework for extrapolating RD treatment effects to a broader class of causal parameters in sharp RD designs. We adopt a partial identification strategy that delivers bounds on the causal parameters of interest. The key identifying assumption is a smoothness restriction, parameterized through sensitivity measures that capture the degree of smoothness of the conditional distribution functions of the potential outcomes on the running variable. Under this framework, we derive sharp bounds on the counterfactual means and distributions of the potential outcomes that are tractable and straightforward to estimate. The proposed approach is well suited to sensitivity analysis in sharp RD designs, and the Monte Carlo simulations provide support for the theoretical results developed in this paper. We further present an empirical application to illus-

trate the practical relevance of these theoretical findings. The results of the empirical application suggest that RD estimates need not be representative of the relevant causal parameters for broader subpopulations or even for the whole population.

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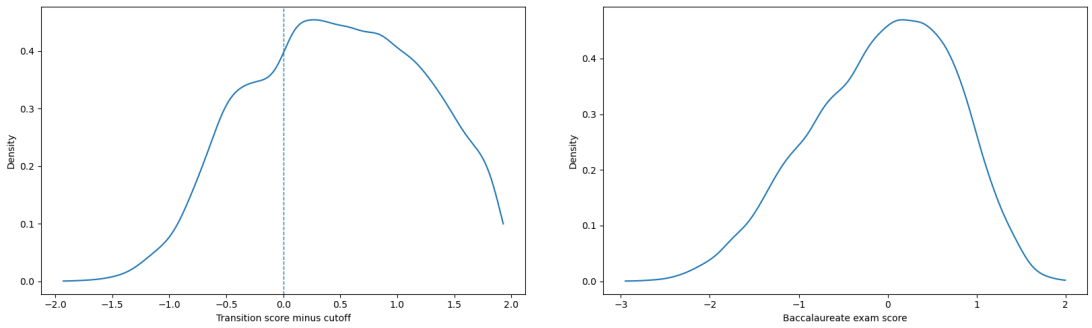
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A Additional Results for the Empirical Application

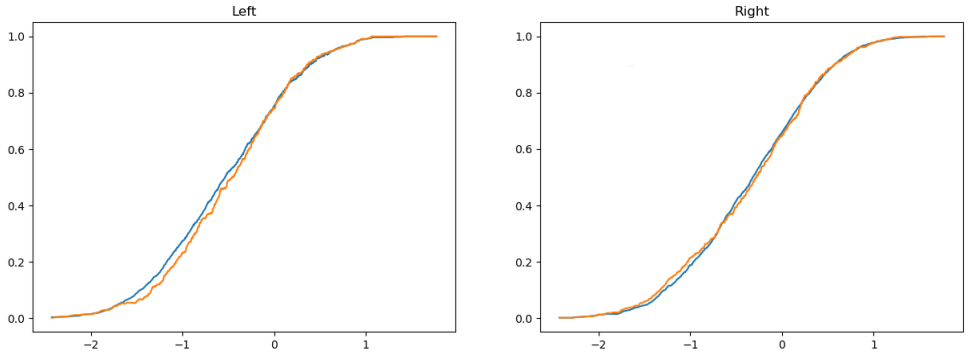
Figure A.1: Kernel Density Estimators of the Outcome and Running Variables



A: Kernel density estimator of R B: Kernel density estimator of Y

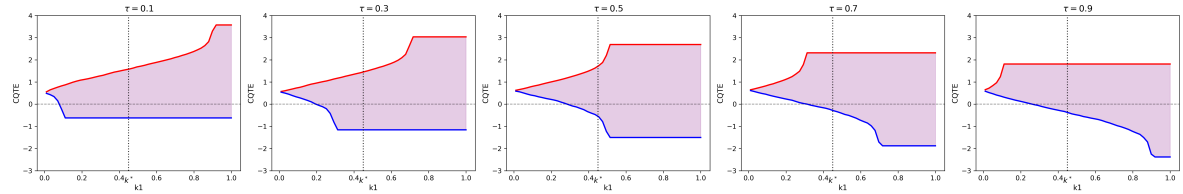
Note: We use the Gaussian kernel for both cases. The bandwidths are set to be 0.110 and 0.117 for R and Y, respectively.

Figure A.2: Comparison of DR and Nonparametric Estimators of $F_{Y_1|R}(\cdot|c)$ and $F_{Y_0|R}(\cdot|c)$

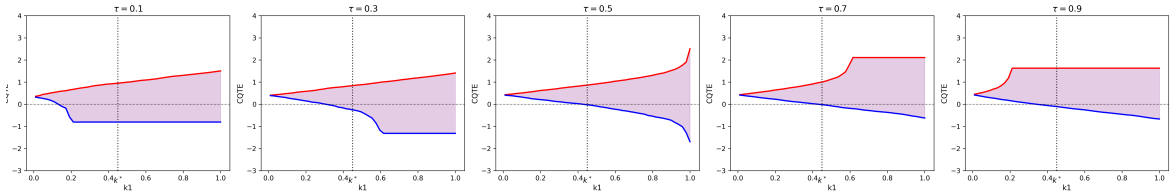


Note: The blue lines are the DR estimators, and the orange lines are the nonparametric local linear estimators of the distribution functions.

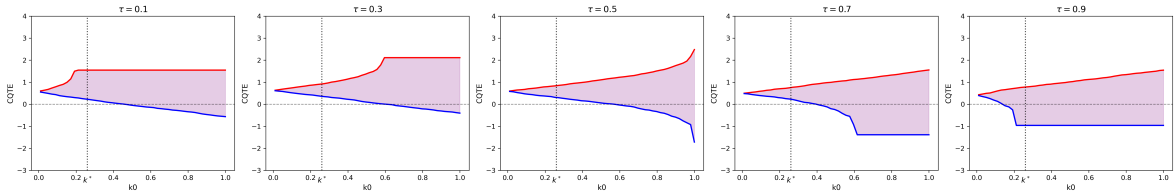
Figure A.3: Sensitivity of Bounds on the CQTE



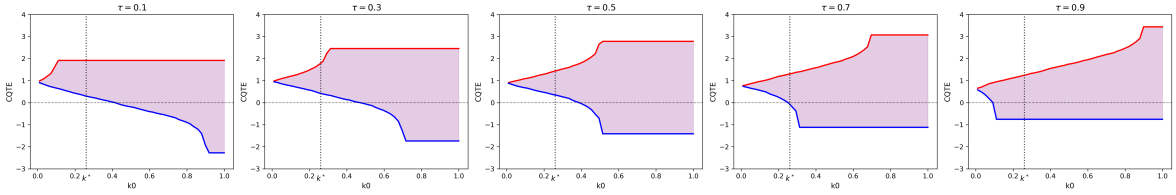
A: CQTE on $R = -1$



B: CQTE on $R = -0.5$



C: CQTE on $R = 0.5$



D: CQTE on $R = 1$

Note: We consider $\tau \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ to estimate the bounds on CQTE. k^* denotes the value of k_0 or k_1, estimated as the maximum derivative of the conditional distribution function over its identified region. Specifically, $k^* = 0.45$ for $R = r < 0$, and $k^* = 0.26$ for $R = r > 0$.

B Mathematical Proofs

B.1 Proof of Theorem 3.1

Proof. We only prove the results on the lower and upper bounds on $E\left[h\left(Y_1\right)|R=r\right]$. The bounds on $E\left[h\left(Y_0\right)|R=r\right]$ can be proven by a similar way.

Let $r, \tilde{r} \in Supp(R)$ be given. Note that under Assumption 2.2, we have

$$E\left[h(Y_1)|R=\tilde{r}\right] - k_{1,h}\left|r-\tilde{r}\right| \leq E\left[h(Y_1)|R=r\right] \leq E\left[h(Y_1)|R=\tilde{r}\right] + k_{1,h}\left|r-\tilde{r}\right|.$$

By Assumption 2.1, $E\left[h(Y_1)|R=\tilde{r}\right]$ is identified for all $\tilde{r} \in \mathcal{R}_1$. Taking the supremum over $\tilde{r} \in \mathcal{R}_1$ to the both sides of $E\left[h(Y_1)|R=\tilde{r}\right] - k_{1,h}\left|r-\tilde{r}\right| \leq E\left[h(Y_1)|R=r\right]$ yields the lower bound. Similarly, the upper bound can be obtained by applying the infimum over $\tilde{r} \in \mathcal{R}_1$ to $E\left[h(Y_1)|R=r\right] \leq E\left[h(Y_1)|R=\tilde{r}\right] + k_{1,h}\left|r-\tilde{r}\right|$. ■

B.2 Proof of Theorem 3.2

Lemma B.1. *Suppose that Assumptions 2.1 and 2.2 hold, and let $d \in \{0,1\}$ be given. Let h be a measurable function such that $E\left[|h(Y_d)|\right] < \infty$. Define*

$$E\left[h(Y_d) \mid R=r_d^\star\right] \equiv \lim_{t \to r_d^\star : t \in \mathcal{R}_d} E\left[h(Y_d) \mid R=t\right].$$

Then, $E\left[h(Y_d) \mid R=r_d^\star\right]$ is well-defined.

Proof. Let $m_{d,h}(r) \equiv E[h(Y_d) \mid R=r]$ for $r \in Supp(R)$. Assumption 2.1 implies that $m_{d,h}(r)$ is identified for $r \in \mathcal{R}_d$. By Assumption 2.2, for any $r, \tilde{r} \in Supp(R)$,

$$|m_{d,h}(r) - m_{d,h}(\tilde{r})| \leq k_{d,h}|r-\tilde{r}|.$$

We first prove existence. For any sequence $(t_n) \subset \mathcal{R}_d$ with $t_n \to r_d^\star$,

$$|m_{d,h}(t_n) - m_{d,h}(t_m)| \leq k_{d,h}|t_n - t_m| \to 0$$

as $n, m \to \infty$, so $(m_{d,h}(t_n))$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, there exists $m^\star \in \mathbb{R}$ such that

$$|m_{d,h}(t_n) - m^\star| \to 0.$$

By the definition above, $E\left[h(Y_d) \mid R = r_d^\star\right] = m^\star$.

We now prove uniqueness. Let $(t_n) \subset \mathcal{R}_d$ and $(s_m) \subset \mathcal{R}_d$ be sequences such that $t_n \to r_d^\star$ and $s_m \to r_d^\star$. By the existence result, there exist $m^\star, \tilde{m}^\star \in \mathbb{R}$ such that $m_{d,h}(t_n) \to m^\star$ and $m_{d,h}(s_m) \to \tilde{m}^\star$. For all $n, m \in \mathbb{N}$,

$$|m^\star - \tilde{m}^\star| \leq |m^\star - m_{d,h}(t_n)| + |m_{d,h}(t_n) - m_{d,h}(s_m)| + |m_{d,h}(s_m) - \tilde{m}^\star|.$$

Taking $\limsup_{n,m\to\infty}$ yields

$$|m^\star - \tilde{m}^\star| \leq \limsup_{n,m\to\infty} |m_{d,h}(t_n) - m_{d,h}(s_m)|.$$

By Assumption 2.2, it follows that

$$|m_{d,h}(t_n) - m_{d,h}(s_m)| \leq k_{d,h}|t_n - s_m|$$

for all $n, m \in \mathbb{N}$. Thus,

$$|m^\star - \tilde{m}^\star| \leq \limsup_{n,m\to\infty} k_{d,h}|t_n - s_m| = 0,$$

since $t_n \to r_d^\star$ and $s_m \to r_d^\star$, implying $m^\star = \tilde{m}^\star$. ■

Proof of Theorem 3.2

Proof. We only prove $LB_1(k_{1,h}, r_0)$ and $UB_1(k_{1,h}, r_0)$, and the other results follow by the same argument. Fix $r_0 \in \mathcal{R}_0$. By Lemma B.1, $E\left[h(Y_1) \mid R = r_1^\star\right]$ is well-defined.

Lower bound. Take any $\tilde{r} \in \mathcal{R}_1$. By Assumption 2.2,

$$E\left[h(Y_1) \mid R = \tilde{r}\right] \leq E\left[h(Y_1) \mid R = r_1^\star\right] + k_{1,h}\left|\tilde{r} - r_1^\star\right|.$$

Since $r_0 < r_1^\star \leq \tilde{r}$, we have $|r_0 - \tilde{r}| = |r_0 - r_1^\star| + |\tilde{r} - r_1^\star|$. Therefore,

$$\begin{aligned} E\left[h(Y_1) \mid R = \tilde{r}\right] - k_{1,h}|r_0 - \tilde{r}| &\leq E[h(Y_1) \mid R = r_1^\star] + k_{1,h}|\tilde{r} - r_1^\star| - k_{1,h}\left(|r_0 - r_1^\star| + |\tilde{r} - r_1^\star|\right) \\ &= E[h(Y_1) \mid R = r_1^\star] - k_{1,h}|r_0 - r_1^\star|, \end{aligned}$$

which implies that $E[h(Y_1) \mid R = r_1^\star] - k_{1,h}\left|r_0 - r_1^\star\right|$ is an upper bound for $E[h(Y_1) \mid R = \tilde{r}] - k_{1,h}\left|r_0 -\right.$

$\tilde{r}\big|$ for all $\tilde{r} \in \mathcal{R}_1$. Taking the supremum over $\tilde{r} \in \mathcal{R}_1$ yields

$$\sup_{\tilde{r}\in\mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] - k_{1,h}|r_0 - \tilde{r}|\} \leq E\left[h(Y_1) \mid R = r_1^\star\right] - k_{1,h}|r_0 - r_1^\star|. \tag{5}$$

To show the reverse inequality, take any sequence $(r_{1n}) \subset \mathcal{R}_1$ such that $r_{1n} \to r_1^\star$. Then, for all $n \in \mathbb{N}$,

$$\sup_{\tilde{r}\in\mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] - k_{1,h}|r_0 - \tilde{r}|\} \geq E\left[h(Y_1) \mid R = r_{1n}\right] - k_{1,h}|r_0 - r_{1n}|.$$

Letting $n \to \infty$ and using Lemma B.1 together with $|r_0 - r_{1n}| \to |r_0 - r_1^\star|$, we obtain

$$\sup_{\tilde{r}\in\mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] - k_{1,h}|r_0 - \tilde{r}|\} \geq E\left[h(Y_1) \mid R = r_1^\star\right] - k_{1,h}|r_0 - r_1^\star|. \tag{6}$$

Combining (5) and (6) yields

$$LB_1(k_{1,h}, r_0) = E\left[h(Y_1) \mid R = r_1^\star\right] - k_{1,h}|r_0 - r_1^\star|.$$

Upper bound. Take any $\tilde{r} \in \mathcal{R}_1$. By Assumption 2.2,

$$E[h(Y_1) \mid R = \tilde{r}] \geq E\left[h(Y_1) \mid R = r_1^\star\right] - k_{1,h}|\tilde{r} - r_1^\star|.$$

Since $r_0 < r_1^\star \leq \tilde{r}$, we have $|r_0 - \tilde{r}| = |r_0 - r_1^\star| + |\tilde{r} - r_1^\star|$. Thus,

$$\begin{aligned} E\left[h(Y_1) \mid R = \tilde{r}\right] + k_{1,h}|r_0 - \tilde{r}| &\geq E\left[h(Y_1) \mid R = r_1^\star\right] - k_{1,h}|\tilde{r} - r_1^\star| + k_{1,h}\left(|r_0 - r_1^\star| + |\tilde{r} - r_1^\star|\right) \\ &= E\left[h(Y_1) \mid R = r_1^\star\right] + k_{1,h}|r_0 - r_1^\star|. \end{aligned}$$

Taking the infimum over $\tilde{r} \in \mathcal{R}_1$ yields

$$\inf_{\tilde{r}\in\mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] + k_{1,h}|r_0 - \tilde{r}|\} \geq E\left[h(Y_1) \mid R = r_1^\star\right] + k_{1,h}|r_0 - r_1^\star|. \tag{7}$$

To show the reverse inequality, take any sequence $(r_{1n}) \subset \mathcal{R}_1$ with $r_{1n} \to r_1^\star$. Then, for all $n \in \mathbb{N}$,

$$\inf_{\tilde{r}\in\mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] + k_{1,h}|r_0 - \tilde{r}|\} \leq E\left[h(Y_1) \mid R = r_{1n}\right] + k_{1,h}|r_0 - r_{1n}|.$$

Letting $n \to \infty$ and using Lemma B.1 together with $|r_0 - r_{1n}| \to |r_0 - r_1^\star|$ yields

$$\inf_{\tilde{r} \in \mathcal{R}_1} \{E\left[h(Y_1) \mid R = \tilde{r}\right] + k_{1,h}|r_0 - \tilde{r}|\} \leq E\left[h(Y_1) \mid R = r_1^\star\right] + k_{1,h}|r_0 - r_1^\star|. \tag{8}$$

Combining (7) and (8) proves the claim. ■

B.3 Proof of Theorem 3.3

We provide two lemmas that will be used for the proof of Theorem 3.3.

Lemma B.2. *Suppose that Assumption 3.1 holds and let $d \in \{0, 1\}$ be given. Define*

$$F_{Y_d|R}(y|r_d^\star) \equiv \lim_{t \to r_d^\star : t \in \mathcal{R}_d} F_{Y_d}(y|t)$$

for each $y \in \mathbb{R}$. Then, $F_{Y_d|R}(\cdot|r_d^\star)$ is a well-defined cdf.

Proof. Let $l^\infty(\mathbb{R})$ denote the space of functions on $\mathbb{R}$ that are measurable and uniformly bounded and define $d_\infty(\cdot, \cdot)$ as the uniform (pseudo-)metric on $l^\infty(\mathbb{R}) \times l^\infty(\mathbb{R})$. That is,

$$l^\infty(\mathcal{Y}) \equiv \left\{ f : \sup_{y \in \mathcal{Y}} |f(y)| < \infty, f \text{ is measurable} \right\}$$

and

$$d_\infty(f, g) \equiv \sup_{y \in \mathcal{Y}} \Big| f(y) - g(y) \Big|.$$

We first prove the existence of $F_{Y_d|R}(\cdot|r_d^\star)$. By Assumption 3.1-(ii), for any $r, \tilde{r} \in Supp(R)$, we have

$$d_\infty\big(F_{Y_d|R}(\cdot \mid r), F_{Y_d|R}(\cdot \mid \tilde{r})\big) \leq k_{d,c}|r - \tilde{r}|.$$

Thus, for any sequence $(t_n) \subset \mathcal{R}_d$ with $t_n \to r_d^\star$,

$$d_\infty\big(F_{Y_d|R}(\cdot \mid t_n), F_{Y_d|R}(\cdot \mid t_m)\big) \leq k_{d,c}|t_n - t_m| \to 0$$

as $n, m \to \infty$, so $\big(F_{Y_d|R}(\cdot \mid t_n)\big)$ is Cauchy in $(l^\infty(\mathbb{R}), d_\infty)$. Since the pseudo-metric space is complete (cf. Theorem 19.1 in Billingsley (1995)), there exists $F^\star \in l^\infty(\mathbb{R})$ such that

$$d_\infty\left(F_{Y_d|R}(\cdot \mid t_n), F^\star(\cdot)\right) \to 0,$$

and it is clear that $F_{Y_d|R}(\cdot|r_d^\star) \equiv F^\star(\cdot)$.

We now consider the uniqueness of $F_{Y_d|R}(\cdot|r_d^\star)$. Let $(t_n) \subset \mathcal{R}_d$ and $(s_m) \subset \mathcal{R}_d$ with $t_n \to r_d^\star$ and $s_m \to r_d^\star$. Assume $F_{Y_d|R}(\cdot \mid t_n) \to F^\star$ and $F_{Y_d|R}(\cdot \mid s_m) \to G^\star$ in d_∞. By the triangle inequality, for all $n, m \in \mathbb{N}$, we have

$$d_\infty\big(F^\star, G^\star\big) \leq d_\infty\left(F^\star, F_{Y_d|R}(\cdot \mid t_n)\right) + d_\infty\left(F_{Y_d|R}(\cdot \mid t_n), F_{Y_d|R}(\cdot \mid s_m)\right) + d_\infty\left(F_{Y_d|R}(\cdot \mid s_m), G^\star\right).$$

Taking $\limsup_{n,m\to\infty}$ yields that

$$d_\infty\big(F^\star, G^\star\big) \leq \limsup_{n,m\to\infty} d_\infty\left(F_{Y_d|R}(\cdot \mid t_n), F_{Y_d|R}(\cdot \mid s_m)\right).$$

By Assumption 3.1-(ii), it follows that

$$d_\infty\left(F_{Y_d|R}(\cdot \mid t_n), F_{Y_d|R}(\cdot \mid s_m)\right) \leq k_{d,c}|t_n - s_m|$$

for all $n, m \in \mathbb{N}$. Thus,

$$\limsup_{n,m\to\infty} d_\infty\left(F_{Y_d|R}(\cdot \mid t_n), F_{Y_d|R}(\cdot \mid s_m)\right) \leq \limsup_{n,m\to\infty} k_{d,c}|t_n - s_m| = 0,$$

since $t_n \to r_d^\star$ and $s_m \to r_d^\star$, implying that $d_\infty(F^\star, G^\star) = 0$.

Lastly, we prove that $F_{Y_d|R}(\cdot|r_d^\star)$ is a proper distribution function. To show that $F_{Y_d|R}(\cdot|r_d^\star)$ is non-decreasing, pick any $y_1 < y_2$. Since for each n, $F_{Y_d|R}(\cdot|t_n)$ is a cdf, and therefore, $F_{Y_d|R}(y_1|t_n) \leq F_{Y_d|R}(y_2|t_n)$. Since $F_{Y_d|R}(\cdot|t_n)$ uniformly converges to $F_{Y_d|R}(\cdot|r_d^\star)$, we have $F_{Y_d|R}(y_1|r_d^\star) \leq F_{Y_d|R}(y_2|r_d^\star)$.

Take any $y \in \mathbb{R}$ and $\epsilon > 0$. By the uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d_\infty\left(F_{Y_d|R}(\cdot|t_N) - F_{Y_d|R}(\cdot|r_d^\star)\right) < \frac{\epsilon}{3}.$$

Since $F_{Y_d|R}(\cdot|t_n)$ is right-continuous for all $n \in \mathbb{N}$, there exists $\delta > 0$ such that for all $h \in (0, \delta)$,

$$\left|F_{Y_d|R}(y + h|t_N) - F_{Y_d|R}(y|t_N)\right| < \frac{\epsilon}{3}.$$

Therefore, for any $h \in (0, \delta)$, we have

$$\begin{aligned}
&\left|F_{Y_d|R}(y+h|r_d^\star) - F_{Y_d|R}(y|r_d^\star)\right| \\
\leq &\left|F_{Y_d|R}(y+h|t_N) - F_{Y_d|R}(y+h|t_n)\right| + \left|F_{Y_d|R}(y+h|t_n) - F_{Y_d|R}(y|t_n)\right| + \left|F_{Y_d|R}(y|t_n) - F_{Y_d|R}(y|r_d^\star)\right| \\
< &\frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon,
\end{aligned}$$

which implies that $F_{Y_d|R}(\cdot|r_d^\star)$ is right-continuous.

Fix $\varepsilon > 0$ and choose N such that

$$d_\infty \left(F_{Y_d|R}(\cdot|t_N) - F_{Y_d|R}(\cdot|r_d^\star)\right) < \frac{\epsilon}{2}.$$

Since $F_{Y_d|R}(\cdot|t_n)$ is a cdf for all $n \in \mathbb{N}$, there exists $M > 0$ such that $F_{Y_d|R}(\cdot|t_N) < \frac{\epsilon}{2}$ for all $y \leq -M$ and $F_{Y_d|R}(\cdot|t_N)) > 1 - \frac{\epsilon}{2}$ for all $y \geq M$. Then, for $y \leq -M$,

$$F_{Y_d|R}(y|r_d^\star) \leq |F_{Y_d|R}(y|r_d^\star) - F_{Y_d|R}(y|t_N)| + F_{Y_d|R}(y|t_N) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

meaning that $\lim_{y \to -\infty} F_{Y_d|R}(y|r_d^\star) = 0$. Similarly, for $y \geq M$,

$$F_{Y_d|R}(y|r_d^\star) \geq F_{Y_d|R}(y|t_N) - |F_{Y_d|R}(y|r_d^\star) - F_{Y_d|R}(y|t_N)| > 1 - \frac{\epsilon}{2} - \frac{\epsilon}{2} = 1 - \epsilon,$$

and thus, $\lim_{y \to +\infty} F_{Y_d|R}(y|r_d^\star) = 1$. In all, $F_{Y_d|R}(\cdot|r_d^\star)$ is well-defined cdf. ■

For given $\delta \in (0, 1)$, define

$$\begin{aligned}
L(u) &\equiv \max\{0, u - \delta\}, \\
U(u) &\equiv \min\{1, u + \delta\},
\end{aligned}$$

as mappings from $[0, 1]$ to $[0, 1]$.

Lemma B.3. *For any $\delta \in (0, 1)$, let $\eta \in \left(0, \min\left\{\delta, \frac{1-\delta}{2}\right\}\right)$ be given. Then, there exist functions $\underline{T}(\cdot; \eta)$ and $\overline{T}(\cdot; \eta)$ from $[0, 1]$ to $[0, 1]$ satisfying the following properties:*

(T1) $\underline{T}(\cdot; \eta)$ and $\overline{T}(\cdot; \eta)$ are continuous and strictly increasing on $[0, 1]$ with $\underline{T}(0; \eta) = \overline{T}(0; \eta) = 0$ and $\underline{T}(1; \eta) = \overline{T}(1; \eta) = 1$;

(T2) for all $u \in [0,1]$*,*

$$L(u) \leq \underline{T}(u;\eta) \leq U(u),$$
$$L(u) \leq \overline{T}(u;\eta) \leq U(u);$$

(T3) for any $\varepsilon \in [0,1]$*, define*

$$T(u;\varepsilon,\eta) \equiv (1-\varepsilon)\underline{T}(u;\eta) + \varepsilon\overline{T}(u;\eta)$$

for all $u \in [0,1]$*. Then,* $T(\cdot;\varepsilon,\eta)$ *is continuous and strictly increasing on* $[0,1]$ *with* $T(0;\varepsilon,\eta) = 0$ *and* $T(1;\varepsilon,\eta) = 1$*, and*

$$L(u) \leq T(u;\varepsilon,\eta) \leq U(u)$$

for all $u \in [0,1]$*;*

(T4) for each fixed $u \in (0,1)$*, as* $\eta \searrow 0$*,*

$$\underline{T}(u;\eta) \searrow L(u),$$
$$\overline{T}(u;\eta) \nearrow U(u).$$

Proof. Fix any $\eta \in \left(0, \min\left\{\delta, \frac{(1-\delta)}{2}\right\}\right)$ and define

$$\alpha_L(\eta) \equiv \delta + \eta,$$
$$\beta_L(\eta) \equiv 1 - \eta,$$

then $0 < \delta < \alpha_L(\eta) < \beta_L(\eta) < 1$. Define $\underline{T}(\cdot;\eta) : [0,1] \to [0,1]$ as

$$\underline{T}(u;\eta) \equiv \begin{cases} \dfrac{L(\alpha_L(\eta))}{\alpha_L(\eta)}u, & 0 \leq u < \alpha_L(\eta), \\ L(u), & \alpha_L(\eta) \leq u \leq \beta_L(\eta), \\ L\big(\beta_L(\eta)\big) + \dfrac{1 - L\big(\beta_L(\eta)\big)}{1-\beta_L(\eta)}\left(u - \beta_L(\eta)\right), & \beta_L(\eta) \leq u \leq 1. \end{cases} \tag{9}$$

Let

$$\alpha_U(\eta) \equiv \eta,$$
$$\beta_U(\eta) \equiv 1 - \delta - \eta,$$

then $0 < \alpha_U(\eta) < \beta_U(\eta) < 1 - \delta < 1$. Define $\overline{T}(\cdot;\eta) : [0,1] \to [0,1]$ as

$$\overline{T}(u;\eta) := \begin{cases} \dfrac{U(\alpha_U(\eta))}{\alpha_U(\eta)} u, & 0 \le u < \alpha_U(\eta), \\ U(u), & \alpha_U(\eta) \le u \le \beta_U(\eta), \\ U\big(\beta_U(\eta)\big) + \dfrac{1 - U\big(\beta_U(\eta)\big)}{1 - \beta_U(\eta)} \big(u - \beta_U(\eta)\big), & \beta_U(\eta) \le u \le 1. \end{cases} \tag{10}$$

We first prove that $\underline{T}(\cdot;\eta)$ satisfies (T1) and (T2). Note that

$$\frac{L(\alpha_L(\eta))}{\alpha_L(\eta)} = \frac{\eta}{\delta + \eta} \in (0,1)$$

and

$$\frac{1 - L\big(\beta_L(\eta)\big)}{1 - \beta_L(\eta)} = \frac{\delta + \eta}{\eta} > 1.$$

Moreover, we have $u > \delta$ and $L(u) = u - \delta$ for $u \in [\alpha_L(\eta), \beta_L(\eta)]$. Hence, each piece is linear with a positive slope and the three pieces match at $u = \alpha_L(\eta)$ and $u = \beta_L(\eta)$; thus $\underline{T}(\cdot;\eta)$ is continuous and strictly increasing on $[0,1]$ with $\underline{T}(0;\eta) = 0$ and $\underline{T}(1;\eta) = 1$. Therefore, (T1) holds for $\underline{T}(\cdot;\eta)$.

We consider verifying that $\underline{T}(\cdot;\eta)$ satisfies (T2). For any $u \in [\alpha_L(\eta), \beta_L(\eta)]$, it is straightforward to see that $\underline{T}(u;\eta) = L(u)$ by construction, so $L(u) \le \underline{T}(u;\eta) \le U(u)$ holds. For $u \in [0, \alpha_L(\eta))$, we have

$$\underline{T}(u;\eta) = \frac{L(\alpha_L(\eta))}{\alpha_L(\eta)} u = \frac{\eta}{\delta + \eta} u,$$

and hence, we obtain

$$\underline{T}(u;\eta) - L(u) = \begin{cases} \dfrac{\eta}{\delta + \eta} u, & 0 \le u \le \delta, \\ \delta \dfrac{\alpha_L(\eta) - u}{\delta + \eta}, & \delta < u < \alpha_L(\eta), \end{cases}$$

which is nonnegative for all $u \in [0, \alpha_L(\eta))$. Therefore, it follows that $L(u) \le \underline{T}(u;\eta)$ for all

$u \in [0, \alpha_L(\eta))$. Moreover, since $\underline{T}(\cdot;\eta)$ and $U(\cdot)$ are increasing, we have, for $u \in [0, \alpha_L(\eta)]$,

$$\underline{T}(u;\eta) \le \underline{T}(\alpha_L(\eta);\eta) = L(\alpha_L(\eta)) = \eta \le \delta = U(0) \le U(u),$$

implying that $\underline{T}(u;\eta) \le U(u)$ for $u \in [0, \alpha_L(\eta))$. Lastly, for $u \in (\beta_L(\eta), 1]$, we have that $L(u) = u - \delta$ and $L\big(\beta_L(\eta)\big) = \beta_L(\eta) - \delta$ since $u > \beta_L(\eta) > \delta$. It then follows that

$$\underline{T}(u;\eta) = L\big(\beta_L(\eta)\big) + \frac{1 - L\big(\beta_L(\eta)\big)}{1 - \beta_L(\eta)}\big(u - \beta_L(\eta)\big),$$

and therefore,

$$\underline{T}(u;\eta) - L(u) = \left(\frac{1 - L\big(\beta_L(\eta)\big)}{1 - \beta_L(\eta)} - 1\right)(u - \beta_L(\eta)) \ge 0$$

since $\frac{1-L(\beta_L(\eta))}{1-\beta_L(\eta)} = \frac{\delta+\eta}{\eta} > 1$ and $u - \beta_L(\eta) > 0$. Thus, $\underline{T}(u;\eta) \ge L(u)$. On the other hand, since $\eta < \delta$, we have $\beta_L(\eta) = 1 - \eta > 1 - \delta$, so for $u \ge \beta_L(\eta)$,

$$U(u) = \min\{1, u + \delta\} = 1,$$

and clearly $\underline{T}(u;\eta) \le 1 = U(u)$. Hence $\underline{T}(u;\eta) \le U(u)$ for all $u \in (\beta_L(\eta), 1]$. In all, we obtain

$$L(u) \le \underline{T}(u;\eta) \le U(u)$$

for all $u \in [0, 1]$.

We now show that $\bar{T}(\cdot;\eta)$ satisfies (T1) and (T2). Note that

$$\frac{U(\alpha_U(\eta))}{\alpha_U(\eta)} = \frac{\alpha_U(\eta) + \delta}{\alpha_U(\eta)} = \frac{\eta + \delta}{\eta} > 1$$

and

$$\frac{1 - U\big(\beta_U(\eta)\big)}{1 - \beta_U(\eta)} = \frac{\eta}{\delta + \eta} \in (0, 1).$$

Moreover, we have $u \le \beta_U(\eta) \le 1 - \delta$ for $u \in [\alpha_U(\eta), \beta_U(\eta)]$, implying that $U(u) = u + \delta$ for $u \in [\alpha_U(\eta), \beta_U(\eta)]$. Hence, each piece is linear with a positive slope, and the three pieces match at $u = \alpha_U(\eta)$ and $u = \beta_U(\eta)$; thus $\overline{T}(\cdot;\eta)$ is continuous and strictly increasing on $[0, 1]$ with $\overline{T}(0;\eta) = 0$ and $\overline{T}(1;\eta) = 1$. Therefore, (T1) holds for $\overline{T}(\cdot;\eta)$.

We verify that $\bar{T}(\cdot;\eta)$ satisfies (T2). For any $u \in [\alpha_U(\eta), \beta_U(\eta)]$, it is clear that $\overline{T}(u;\eta) = U(u)$ by construction, and thus, we have $L(u) \le \overline{T}(u;\eta) \le U(u)$ for all $u \in [\alpha_U(\eta), \beta_U(\eta)]$. For $u \in$

$[0, \alpha_U(\eta))$, we have $L(u) = 0$ since $u < \alpha_U(\eta) = \eta < \delta$. In addition, it follows that

$$\overline{T}(u;\eta) = \left(1 + \frac{\delta}{\eta}\right) u,$$

and therefore,

$$\overline{T}(u;\eta) - U(u) = \left(1 + \frac{\delta}{\eta}\right) u - (u + \delta)) = \frac{\delta}{\eta}(u - \eta) \leq 0$$

because $0 \leq u < \eta$. Consequently, we obtain $L(u) \leq \overline{T}(u;\eta) \leq U(u)$ for $u \in [0, \alpha_U(\eta))$. Lastly, for $u \in (\beta_U(\eta), 1]$, it can be easily shown that

$$\overline{T}(u;\eta) = 1 - \eta + \frac{\eta}{\delta + \eta}(u - \beta_U(\eta)).$$

Since $\eta < \delta$, we have

$$\overline{T}\big(\beta_U(\eta);\eta\big) = 1 - \eta > 1 - \delta = L(1).$$

Because $L(\cdot)$ and $\overline{T}(\cdot;\eta)$ are nondecreasing on $[0,1]$, it follows that, for all $u \in (\beta_U(\eta), 1]$,

$$L(u) \leq L(1) < \overline{T}\big(\beta_U(\eta);\eta\big) \leq \overline{T}(u;\eta).$$

In consequence, we have $L(u) \leq \overline{T}(u;\eta)$ for all $u \in (\beta_U(\eta), 1]$. Note that $U(u) = u + \delta$ on $u \in (\beta_U(\eta), 1]$ and $U(u) \equiv 1$ on $[1 - \delta, 1]$. Therefore, we obtain

$$U(u) - \overline{T}(u;\eta) = \begin{cases} \dfrac{\delta}{\delta + \eta}\big(\delta + \eta + u - 1\big), & \beta_U(\eta) < u \leq 1 - \delta, \\ \dfrac{\eta}{\delta + \eta}(1 - u), & 1 - \delta < u \leq 1, \end{cases}$$

and $U(u) - \overline{T}(u;\eta) \geq 0$ or all $u \in (\beta_U(\eta), 1]$. In all, we have

$$L(u) \leq \overline{T}(u;\eta) \leq U(u)$$

for all $u \in [0,1]$. Therefore, $\underline{T}(\cdot;\eta)$ and $\overline{T}(\cdot;\eta)$ satisfy (T1) and (T2).

We now show that (T3) holds. Let $\varepsilon \in [0,1]$ be given. For $0 \leq u_1 < u_2 \leq 1$, we have

$$\begin{aligned} T(u_2;\varepsilon,\eta) - T(u_1;\varepsilon,\eta) &= (1 - \varepsilon)\left\{\underline{T}(u_2;\eta) - \underline{T}(u_1;\eta)\right\} \\ &\quad + \varepsilon\left\{\overline{T}(u_2;\eta_d) - \overline{T}(u_1;\eta_d)\right\} > 0 \end{aligned}$$

since $\underline{T}(\cdot;\eta)$ and $\bar{T}(\cdot;\eta)$ are strictly increasing on $[0,1]$ by (T1). Therefore, $T(\cdot;\varepsilon,\eta)$ is strictly

increasing on $[0,1]$. Similarly, (T1) implies that $T(0;\varepsilon,\eta) = 0$ and $T(1;\varepsilon,\eta) = 1$. Lastly, since $T(u;\varepsilon,\eta)$ is a convex combination of $\underline{T}(u;\eta)$ and $\overline{T}(u;\eta)$, we obtain

$$L(u) \leq T(u;\varepsilon,\eta) \leq U(u)$$

for all $u \in [0,1]$ from (T2).

We consider (T4) with $\underline{T}(u;\eta)$. Fix $u \in (0,1)$ and suppose that $0 < u \leq \delta$. Then, for any $\eta > 0$, we have $u \leq \delta < \alpha_L(\eta)$, and thus,

$$\underline{T}(u;\eta) = \frac{L(\alpha_L(\eta))}{\alpha_L(\eta)}u = \frac{\eta}{\delta+\eta}u.$$

For any $\eta^{'}$ such that $0 < \eta^{'} < \eta$, it then follows that

$$\underline{T}\left(u;\eta^{'}\right) - \underline{T}\left(u;\eta\right) = u\frac{\delta\left(\eta^{'}-\eta\right)}{(\delta+\eta)\left(\delta+\eta^{'}\right)} < 0.$$

Since $\eta/(\delta+\eta) \to 0$ as $\eta \searrow 0$, we obtain $\underline{T}(u;\eta) \searrow 0 = L(u)$ as $\eta \searrow 0$ for $u \in (0,\delta]$.

Suppose that $\delta < u < 1$. Choose $\bar{\eta}(u) > 0$ such that $\delta + \bar{\eta}(u) < u < 1 - \bar{\eta}(u)$. Then, for all $0 < \eta < \bar{\eta}(u)$, we have $\alpha_L(\eta) = \delta + \eta < u < 1 - \eta = \beta_L(\eta)$, and thus,

$$\underline{T}(u;\eta) = L(u) = u - \delta$$

for all η such that $0 < \eta < \bar{\eta}(u)$ and $u \in (\delta, 1)$. In all, $\underline{T}(u;\eta) \searrow L(u)$ as $\eta \searrow 0$ for all $u \in (0,1)$.

We now consider (T4) with $\overline{T}(u;\eta)$. Fix $u \in (0,1)$ and suppose that $0 < u < 1 - \delta$. Choose $\bar{\eta}(u) < u < 1 - \delta - \bar{\eta}(u)$. Then, for all $0 < \eta < \bar{\eta}(u)$ we have $\alpha_U(\eta) = \eta < u < 1 - \delta - \eta = \beta_U(\eta)$, and thus,

$$\overline{T}(u;\eta) = U(u)$$

for all $0 < \eta < \bar{\eta}(u)$ and $0 < u < 1 - \delta$.

Suppose that $1 - \delta \leq u < 1$. Then, for any $\eta > 0$, we have $\beta_U(\eta) = 1 - \delta_d - \eta < 1 - \delta \leq u$, and thus,

$$\overline{T}(u;\eta) = U\big(\beta_U(\eta)\big) + \frac{1 - U\big(\beta_U(\eta)\big)}{1 - \beta_U(\eta)}\big(u - \beta_U(\eta)\big) = 1 + \frac{\eta(u-1)}{\delta+\eta}.$$

Take any $\eta^{'}$ such that $0 < \eta^{'} < \eta$. Then,

$$\overline{T}\left(u;\eta^{'}\right) - \overline{T}\left(u;\eta\right) = (u-1)\frac{\delta(\eta^{'}-\eta)}{(\delta+\eta)(\delta+\eta^{'})} > 0$$

since $(u-1)<0$ and $\left(\eta^{'}-\eta\right)<0$. Since $\eta(u-1)/(\delta+\eta)\to 0$ as $\eta\searrow 0$, $\overline{T}(u;\eta)\nearrow 1=U(u)$ as $\eta\searrow 0$ for $u\in[1-\delta,1)$. As a result, $\overline{T}(u;\eta)\nearrow U(u)$ as $\eta\searrow 0$ for all $u\in(0,1)$.

This establishes that (T4) holds with $\underline{T}(u;\eta)$ and $\overline{T}(u;\eta)$ defined in (9) and (10). ■

Proof of Theorem 3.3

Proof. **Proof of Claim (a)**: We show that $F_{Y_d|R}(\cdot\mid r_{d'})\in\mathcal{F}^{k_{d,c}}_{Y_d|r_{d'}}$ for given $d\in\{0,1\}$ and $r_{d'}\in\mathcal{R}_{d'}$, where $d^{'}=1-d$. For any $\tilde{r}\in\mathcal{R}_d$ and $y\in\mathbb{R}$, we have

$$\left|F_{Y_d|R}(y\mid r_{d'})-F_{Y_d|R}(y\mid\tilde{r})\right|\leq k_{d,c}|r_{d'}-\tilde{r}|$$

by Assumption 3.1. It follows that

$$F_{Y_d|R}(y\mid\tilde{r})-k_{d,c}|r_{d'}-\tilde{r}|\leq F_{Y_d|R}(y\mid r_{d'})\leq F_{Y_d|R}(y\mid\tilde{r})+k_{d,c}|r_{d'}-\tilde{r}|.$$

Taking the supremum and infimum over $\tilde{r}\in\mathcal{R}_d$ yields

$$\sup_{\tilde{r}\in\mathcal{R}_d}\left\{F_{Y_d|R}(y\mid\tilde{r})-k_{d,c}|r_{d'}-\tilde{r}|\right\}\leq F_{Y_d|R}(y\mid r_{d'})\leq\inf_{\tilde{r}\in\mathcal{R}_d}\left\{F_{Y_d|R}(y\mid\tilde{r})+k_{d,c}|r_{d'}-\tilde{r}|\right\}. \tag{11}$$

Since $F_{Y_d|R}(y\mid r^{\star}_d)$ is well defined by Lemma B.2 for each $d\in\{0,1\}$, we have

$$\sup_{\tilde{r}\in\mathcal{R}_d}\left\{F_{Y_d|R}(y\mid\tilde{r})-k_{d,c}|r_{d'}-\tilde{r}|\right\}=F_{Y_d|R}(y\mid r^{\star}_d)-k_{d,c}|r_{d'}-r^{\star}_d|$$

and

$$\inf_{\tilde{r}\in\mathcal{R}_d}\left\{F_{Y_d|R}(y\mid\tilde{r})+k_{d,c}|r_{d'}-\tilde{r}|\right\}=F_{Y_d|R}(y\mid r^{\star}_d)+k_{d,c}|r_{d'}-r^{\star}_d|,$$

by Theorem 3.2. Invoking the logical bounds on the cdf, we complete the proof.

Proof of Claim (b): For $d\in\{0,1\}$, let $d^{'}\equiv 1-d$ and $\delta_d(r_{d'})\equiv k_{d,c}|r_{d'}-r^{\star}_d|$. If $\delta_d(r_{d'})=0$, then

$$\underline{F}^{k_{d,c}}_{Y_d|R}(\cdot\mid r_{d'})=\overline{F}^{k_{d,c}}_{Y_d|R}(\cdot\mid r_{d'})=F_{Y_d|R}(\cdot\mid r^{\star}_d),$$

and no further construction is required. If $\delta_d(r_{d'})\geq 1$, then

$$\underline{F}^{k_{d,c}}_{Y_d|R}(\cdot\mid r_{d'})\equiv 0,\overline{F}^{k_{d,c}}_{Y_d|R}(\cdot\mid r_{d'})\equiv 1,$$

and there is nothing to prove. Therefore, we focus on the case that $\delta_d(r_{d'})\in(0,1)$.

Note that

$$\underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) = L_d\left(F_{Y_d|R}(y \mid r^\star_d)\right)$$

and

$$\overline{F}^{k_d}_{Y_d|R}(y \mid r_{d'}) = U_d\left(F_{Y_d|R}(y \mid r^\star_d)\right),$$

where $L_d, U_d : [0,1] \to [0,1]$ are defined by

$$L_d(u) \equiv \max\{0, u - \delta_d(r_{d'})\},$$
$$U_d(u) \equiv \min\{1, u + \delta_d(r_{d'})\}.$$

Let $\eta_d \in \left(0, \min\left\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\right\}\right)$ be given and define $\underline{T}^{k_{d,c}}(\cdot;\eta_d)$ and $\overline{T}^{k_{d,c}}(\cdot;\eta_d)$ on $[0,1]$ as

$$\underline{T}^{k_{d,c}}(u;\eta_d) \equiv \begin{cases} \dfrac{L_d(\alpha_L(\eta_d))}{\alpha_L(\eta_d)}u, & 0 \le u < \alpha_L(\eta_d), \\ L_d(u), & \alpha_L(\eta_d) \le u \le \beta_L(\eta_d), \\ L_d\big(\beta_L(\eta_d)\big) + \dfrac{1 - L_d\big(\beta_L(\eta_d)\big)}{1 - \beta_L(\eta_d)}\big(u - \beta_L(\eta_d)\big), & \beta_L(\eta_d) \le u \le 1, \end{cases}$$

and

$$\overline{T}^{k_{d,c}}(u;\eta_d) \equiv \begin{cases} \dfrac{U_d(\alpha_U(\eta_d))}{\alpha_U(\eta_d)}u, & 0 \le u < \alpha_U(\eta_d), \\ U_d(u), & \alpha_U(\eta_d) \le u \le \beta_U(\eta_d), \\ U_d\big(\beta_U(\eta_d)\big) + \dfrac{1 - U_d\big(\beta_U(\eta_d)\big)}{1 - \beta_U(\eta_d)}\big(u - \beta_U(\eta_d)\big), & \beta_U(\eta_d) \le u \le 1, \end{cases}$$

where $\alpha_L(\eta_d) \equiv \delta_d(r_{d'}) + \eta_d$, $\beta_L(\eta_d) \equiv 1 - \eta_d$, $\alpha_U(\eta_d) \equiv \eta_d$, and $\beta_U(\eta_d) := 1 - \delta_d - \eta_d$. Then, Lemma B.3 implies that $\underline{T}^{k_{d,c}}(\cdot;\eta_d)$ and $\overline{T}^{k_{d,c}}(\cdot;\eta_d)$ satisfy the following properties:

(Td1) $\underline{T}^{k_{d,c}}(\cdot;\eta_d)$ and $\overline{T}^{k_{d,c}}(\cdot;\eta_d)$ are continuous and strictly increasing on $[0,1]$ with $\underline{T}^{k_{d,c}}(0;\eta_d) = \overline{T}^{k_{d,c}}(0;\eta_d) = 0$ and $\underline{T}^{k_{d,c}}(1;\eta_d) = \overline{T}^{k_{d,c}}(1;\eta_d) = 1$;

(Td2) for all $u \in [0,1]$,

$$L_d(u) \le \underline{T}^{k_{d,c}}(u;\eta_d) \le U_d(u),$$
$$L_d(u) \le \overline{T}^{k_{d,c}}(u;\eta_d) \le U_d(u);$$

(Td3) for any $\varepsilon_d \in [0,1]$, define

$$T^{k_{d,c}}(u;\varepsilon_d,\eta_d) \equiv (1-\varepsilon_d)\underline{T}^{k_{d,c}}(u;\eta_d) + \varepsilon_d \overline{T}^{k_{d,c}}(u;\eta_d) \tag{12}$$

for all $u \in [0,1]$. Then, $T^{k_{d,c}}(\cdot;\varepsilon_d,\eta_d)$ is continuous and strictly increasing on $[0,1]$ with $T^{k_{d,c}}(0;\varepsilon_d,\eta_d) = 0$ and $T^{k_{d,c}}(1;\varepsilon_d,\eta_d) = 1$, and

$$L_d(u) \leq T^{k_{d,c}}(u;\varepsilon_d,\eta_d) \leq U_d(u) \quad \text{for all } u \in [0,1];$$

(Td4) for each fixed $u \in (0,1)$, as $\eta_d \searrow 0$,

$$\underline{T}^{k_{d,c}}(u;\eta_d) \searrow L_d(u)$$

and

$$\overline{T}^{k_{d,c}}(u;\eta_d) \nearrow U_d(u).$$

Let $\epsilon_d \in [0,1]$ be given and define

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d) \equiv T^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d);\varepsilon_d,\eta_d\big). \tag{13}$$

We will proceed with showing that (1) for each $\epsilon_d \in [0,1]$ and $\eta_d \in \left(0, \min\left\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\right\}\right)$, $F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d)$ is a valid cdf and lies in the cdf band at $R = r^\star_d$; (2) for each $y \in \mathbb{R}$ and $\eta_d \in \left(0, \min\left\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\right\}\right)$, $F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d)$ is continuous in ϵ_d on $[0,1]$.

Note that $F_{Y_d|R}(\cdot \mid r^\star_d)$ is a cdf by Lemma B.2. It also follows from (Td3) that for any fixed (ε_d,η_d) the map $T^{k_{d,c}}(\cdot;\varepsilon_d,\eta_d)$ is continuous and strictly increasing from $[0,1]$ to $[0,1]$ such that $T^{k_{d,c}}(0;\varepsilon_d,\eta_d) = 0$ and $T^{k_d}(1;\varepsilon_d,\eta_d) = 1$. Therefore, $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'};\varepsilon_d,\eta_d) = T^{k_{d,c}}\big(F_{Y_d|R}(\cdot \mid r^\star_d);\varepsilon_d,\eta_d\big)$ is a valid cdf. Moreover, by (Td3),

$$L_d(u) \leq T^{k_d}(u;\varepsilon_d,\eta_d) \leq U_d(u)$$

for all $u \in [0,1]$. Therefore, substituting $u = F_{Y_d|R}(y \mid r^\star_d)$ yields that $\underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) = L_d\big(F_{Y_d|R}(y \mid r^\star_d)\big)$ and $\overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) = U_d\big(F_{Y_d|R}(y \mid r^\star_d)\big)$. Combining these together, we obtain that, for all $y \in \mathbb{R}$,

$$\underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \leq F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d) \leq \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}),$$

implying that $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}^{k_{d,c}}_{Y_d|r_{d'}}$.

We now show that $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$ satisfies (i), (ii) and (iii) in Claim (b) of the theorem.

Let $F_R(\cdot)$ denote the distribution function of R. Since $F_R(\cdot)$ is identified from data, it is enough to show that $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$ for $d \in \{0,1\}$ can be jointly obtained. To this end, we construct conditional cdfs $\widetilde{F}_{Y_d|R}(\cdot \mid r)$ for all $r \in Supp(R)$ and $d \in \{0,1\}$ such that:

(S1) they coincide with $F^{k_d}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$ at $r_{d'} \in \mathcal{R}_{d'}$;

(S2) for each $r \in Supp(R)$ and $d \in \{0,1\}$, $\tilde{F}_{Y_d|R}(\cdot|r)$ is a cdf;

(S3) they satisfy Assumption 3.1.

Fix $d \in \{0,1\}$ and let $d' \equiv 1-d$. By construction, $\mathcal{R}_d \cap \mathcal{R}_{d'} = \varnothing$ and $\mathcal{R}_d \cup \mathcal{R}_{d'} = Supp(R)$. Let $r^\star_d$ denote the boundary point on side d; by Lemma A.1, $F_{Y_d|R}(\cdot \mid r^\star_d)$ is well-defined and a c.d.f. Let $\Delta_d := |r_{d'} - r^\star_d|$ and $\delta_d \equiv k_{d,c}\Delta_d$ as in Part 2. Define the weight function $\tau_d : \mathcal{R}_{d'} \to [0,1]$ by

$$\tau_d(r) := \min\left\{ \frac{|r - r^\star_d|}{\Delta_d},\, 1 \right\}$$

for $r_d \in \mathcal{R}_d$. Fix $\varepsilon_d \in [0,1]$ and $\eta_d \in \big(0, \min\{\delta_d, (1-\delta_d)/2\}\big)$. For $r \in Supp(R)$ and $y \in \mathbb{R}$, define

$$\widetilde{F}_{Y_d|R}(y \mid r; \varepsilon_d, \eta_d) \equiv \begin{cases} F_{Y|R,D=d}(y \mid r), & r \in \mathcal{R}_d, \\ (1-\tau_d(r))\, F_{Y_d|R}(y \mid r^\star_d) + \tau_d(r)\, F^{k_d}_{Y_d|R}\big(y \mid r_{d'}; \varepsilon_d, \eta_d\big), & r \in \mathcal{R}_{d'}. \end{cases} \tag{14}$$

Since $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d) \in \mathcal{F}^{k_{d,c}}_{Y_d|r_{d'}}$, we have

$$\sup_{y \in \mathbb{R}} \big| F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y \mid r^\star_d) \big| \le \delta_d = k_d \Delta_d. \tag{15}$$

We now verify properties 1–3 above for this choice. For convenience, write $\tilde{F}_{Y_d|R}(\cdot \mid r) := \widetilde{F}_{Y_d|R}(\cdot \mid r; \varepsilon_d, \eta_d)$ in the remainder of this part.

1. Since $r_{d'} \in \mathcal{R}_{d'}$ and $\tau_d(r_{d'}) = 1$, we have $\widetilde{F}_{Y_d|R}(\cdot \mid r_{d'}) = F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$.

2. If $r \in \mathcal{R}_d$, then by (14) we have $\widetilde{F}_{Y_d|R}(\cdot \mid r) = F_{Y|R,D=d}(\cdot \mid r)$, a conditional distribution in y and thus a c.d.f. If $r \in \mathcal{R}_{d'}$, then $\widetilde{F}_{Y_d|R}(\cdot \mid r)$ is a convex combination (with the weight $\tau_d(r) \in [0,1]$) of the cdf $F_{Y_d|R}(\cdot \mid r^\star_d)$ and $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$. Convex combinations preserve monotonicity, right–continuity, and the limits 0 and 1; hence $\widetilde{F}_{Y_d|R}(\cdot \mid r)$ is a cdf for every $r \in Supp(R)$.

3. By construction, on $\mathcal{R}_d$ we have $\widetilde{F}_{Y_d|R}(\cdot \mid r) = F_{Y|R,D=d}(\cdot \mid r)$, so Assumption DI holds. For Assumption 3.1, we consider three cases.

Case 1: $r, r' \in \mathcal{R}_d$. For any $y \in \mathbb{R}$,

$$\begin{aligned}\left|\widetilde{F}_{Y_d|R}(y \mid r) - \widetilde{F}_{Y_d|R}(y \mid r')\right| &= \left|F_{Y|R,D=d}(y \mid r) - F_{Y|R,D=d}(y \mid r')\right| \\ &= \left|F_{Y_d|R}(y \mid r) - F_{Y_d|R}(y \mid r')\right| \\ &\leq k_{d,c}|r - r'|,\end{aligned}$$

where the last inequality follows from Assumption DS for $F_{Y_d|R}$.

Case 2: $r, r' \in \mathcal{R}_{d'}$. For any $y \in \mathbb{R}$,

$$\begin{aligned}\left|\widetilde{F}_{Y_d|R}(y \mid r) - \widetilde{F}_{Y_d|R}(y \mid r')\right| &= \left|\tau_d(r) - \tau_d(r')\right| \cdot \left|F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y \mid r^\star_d)\right| \\ &\leq \frac{|r - r'|}{\Delta_d} k_{d,c}\Delta_d = k_{d,c}|r - r'|,\end{aligned}$$

where the inequality uses $|\tau_d(r) - \tau_d(r')| \leq |r - r'|/\Delta_d$ and (15).

Case 3: $r \in \mathcal{R}_d$, $r' \in \mathcal{R}_{d'}$. For any $y \in \mathbb{R}$,

$$\begin{aligned}\left|\widetilde{F}_{Y_d|R}(y \mid r') - \widetilde{F}_{Y_d|R}(y \mid r)\right| &\leq \left|\widetilde{F}_{Y_d|R}(y \mid r') - F_{Y_d|R}(y \mid r^\star_d)\right| + \left|F_{Y_d|R}(y \mid r^\star_d) - \widetilde{F}_{Y_d|R}(y \mid r)\right| \\ &= \tau_d(r') \cdot \left|F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y \mid r^\star_d)\right| + \left|F_{Y_d|R}(y \mid r^\star_d) - F_{Y|R,D=d}(y \mid r)\right| \\ &= \tau_d(r')\left|F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) - F_{Y_d|R}(y \mid r^\star_d)\right| + \left|F_{Y_d|R}(y \mid r^\star_d) - F_{Y_d|R}(y \mid r)\right| \\ &\leq \frac{|r' - r^\star_d|}{\Delta_d} k_{d,c}\Delta_d + k_{d,c}|r - r^\star_d| \\ &= k_{d,c}|r' - r|,\end{aligned}$$

where the first line is the triangle inequality, the second line uses (14), the third line replaces $F_{Y|R,D=d}$ by $F_{Y_d|R}$, the fourth line uses $|\tau_d(r')| \leq |r' - r^\star_d|/\Delta_d$, (15), and Assumption 3.1 for $F_{Y_d|R}$, and the last equality uses that $r^\star_d$ lies between r and r'.

We turn to prove that $F^{k_{d,c}}_{Y_d|R}(\cdot \mid r_{d'}; \varepsilon_d, \eta_d)$ satisfies (ii). Fix $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid r^\star_d) < 1$ and set $u^\star_d \equiv F_{Y_d|R}(y \mid r^\star_d) \in (0, 1)$. Note that

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) = T^{k_{d,c}}\left(u^\star; \varepsilon_d, \eta_d\right)$$

and

$$L_d(u^\star) = \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}),$$
$$U_d(u^\star) = \overline{F}^{k_d}_{Y_d|R}(y \mid r_{d'}).$$

Applying the monotone convergence leads to that, as $\eta_d \searrow 0$,

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 0, \eta_d) = \underline{T}^{k_{d,c}}(u^\star; \eta_d) \searrow L_d(u^\star) = \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'})$$

and

$$F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 1, \eta_d) = \overline{T}^{k_{d,c}}(u^\star; \eta_d) \nearrow U_d(u^\star) = \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}).$$

Lastly, observe that $T^{k_{d,c}}(u; \varepsilon_d, \eta_d) = (1-\varepsilon_d)\underline{T}^{k_{d,c}}(u; \eta_d) + \varepsilon_d \overline{T}^{k_{d,c}}(u; \eta_d)$ for fixed $y \in \mathbb{R}$ and η_d. Therefore,

$$\begin{aligned} F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d) &= T^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d); \varepsilon_d, \eta_d\big) \\ &= (1-\varepsilon_d)\underline{T}^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d); \eta_d\big) + \varepsilon_d \overline{T}^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d); \eta_d\big). \end{aligned}$$

Since $(1-\varepsilon_d)\underline{T}^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d); \eta_d\big) + \varepsilon_d \overline{T}^{k_{d,c}}\big(F_{Y_d|R}(y \mid r^\star_d); \eta_d\big)$ is a linear function in ϵ_d, $F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous in ϵ_d on $[0,1]$.

Proof of Claim (c): We apply the above results to obtain pointwise bounds at the target point $r_{d'}$. Fix $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid r^\star_d) < 1$ and let $e \in \Big(\underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}), \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'})\Big)$. By (ii) in Claim (b), there exists $\eta^*_d > 0$ such that $e \in \Big[F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 0, \eta^*_d), F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; 1, \eta^*_d)\Big]$. Since the map $\varepsilon_d \mapsto F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon_d, \eta^*_d)$ is continuous on $[0,1]$ by (iii) in Claim (b), there exists $\varepsilon^*_d \in [0,1]$ such that $e = F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}; \varepsilon^*_d, \eta^*_d)$ by the intermediate value theorem. Thus, the value e is attainable. ■

B.4 Proof of Theorem 3.5

Let $\tau \in (0,1)$ and $\epsilon \in [0,1]$ be given. Fix $d \in \{0,1\}$ and $\eta_d \in \Big(0, \min\Big\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\Big\}\Big)$ with $d' = 1-d$ and $\delta_d(r_{d'}) \equiv k_{d,c}|r_{d'} - r^\star_d|$. Define $\big(T^{k_{d,c}}\big)^{-1}(\tau, \epsilon, \eta_d)$ as the value u such that $T^{k_{d,c}}(u, \epsilon, \eta_d) = \tau$, where the definition of $T^{k_{d,c}}(u, \epsilon, \eta_d)$ is in equation (12).

Lemma B.4. *Suppose that Assumptions 3.1 and 3.2 hold. Fix $d \in \{0,1\}$ and $\eta_d \in \Big(0, \min\Big\{\delta_d(r_{d'}), \frac{1-\delta_d(r_{d'})}{2}\Big\}\Big)$. Then, for each $\tau \in (0,1)$, the map $\epsilon \mapsto \big(T^{k_{d,c}}\big)^{-1}(\tau, \epsilon, \eta_d)$ is continuous on $[0,1]$.*

Proof. For simplicity of notation, let $T^{-1}(\tau;\varepsilon) = \left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$ for a given value of η_d. Note that for each $\tau \in (0,1)$, the value of $\left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$ exists and is unique since $T^{k_{d,c}}(\cdot,\epsilon,\eta_d)$ is continuous and strictly increasing from the proof of Theorem 3.3.

Let (ε_n) be any sequence in $[0,1]$ with $\varepsilon_n \to \varepsilon$ as $n \to \infty$. For each n, define

$$u_n \equiv \left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon_n,\eta_d) \in [0,1],$$

so that, by definition of $\left(T^{k_{d,c}}\right)^{-1}$,

$$T^{k_{d,c}}(u_n,\epsilon_n,\eta_d) = \tau$$

for all $n \in \mathbb{N}$. Since (u_n) is a sequence in the compact set $[0,1]$, there exists a subsequence (u_{n_m}) such that $u_{n_m} \to u^\star$ for some $u^\star \in [0,1]$ as $m \to \infty$. Because $\varepsilon_{n_m} \to \varepsilon$ and $T^{k_{d,c}}(\cdot,\cdot\eta_d)$ is continuous on $[0,1]\times[0,1]$, we obtain

$$T^{k_{d,c}}(u^\star,\epsilon,\eta_d) = \lim_{k\to\infty} T^{k_{d,c}}(u_{n_m},\varepsilon_{n_m},\eta_d) = \lim_{m\to\infty}\tau = \tau.$$

For this fixed ε, the map $u \mapsto T^{k_{d,c}}(u,\epsilon,\eta_d)$ is strictly increasing and continuous on $[0,1]$, with $T^{k_{d,c}}(0,\epsilon,\eta_d) = 0$ and $T^{k_{d,c}}(1,\epsilon,\eta_d) = 1$ from the proof of Theorem 3.3. Hence, the equation $T^{k_{d,c}}(u,\epsilon,\eta_d) = \tau$ has a unique solution in $[0,1]$, namely $u^\star = \left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$. Thus, every convergent subsequence of (u_n) has the limit $\left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$. Hence (u_n) converges to $\left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$, which implies that

$$\left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon_n,\eta_d) \to \left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$$

as $n \to \infty$. Since (ϵ_n) was arbitrary, the map $\epsilon \mapsto \left(T^{k_{d,c}}\right)^{-1}(\tau,\epsilon,\eta_d)$ is continuous on $[0,1]$. ■

Let

$$Q^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'};\varepsilon_d,\eta_d) \equiv \inf\left\{y \in \mathbb{R} : F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d) \geq \tau\right\},$$

where the definition of $F^{k_{d,c}}_{Y_d|R}(y \mid r_{d'};\varepsilon_d,\eta_d)$ is in (13).

Lemma B.5. *Suppose that Assumptions 3.1 and 3.2 hold. Then, the map $\varepsilon_d \mapsto Q^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'};\varepsilon_d,\eta_d)$ is continuous on $[0,1]$.*

Pick any $\tau \in (0,1)$ and any sequence (η_n) with $0 < \eta_n < \min\{\delta_d(r_{d'}),(1-\delta_d(r_{d'}))/2\}$ for all n and $\eta_n \searrow 0$. Note that, from the proof of Theorem 3.3 that, for each $y \in \mathbb{R}$ with $0 < F_{Y_d|R}(y \mid$

$r_d^\star) < 1$, we have

$$F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 1, \eta_d) \nearrow \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}),$$
$$F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 0, \eta_d) \searrow \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}).$$

as $\eta_d \searrow 0$. Therefore, the sequence of sets $\mathcal{Y}_{1n} \equiv \left\{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 1, \eta_n) \geq \tau\right\}$ is monotonically increasing. In consequence, the sequence $y_{1n} \equiv \inf \mathcal{Y}_{1n}$ is monotonically decreasing and converges to

$$\inf\left\{y \in \mathbb{R} : \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} = \underline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}),$$

which may be $-\infty$. Next, we observe that the sequence of sets $\mathcal{Y}_{0n} \equiv \left\{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; 0, \eta_n) \geq \tau\right\}$ is monotonically decreasing, implying that the sequence $y_{0n} \equiv \inf \mathcal{Y}_{0n}$ is monotonically increasing and converges to

$$\inf\left\{y \in \mathbb{R} : \underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \geq \tau\right\} = \overline{Q}_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}),$$

which may be $+\infty$. Note that

$$\begin{aligned}
Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d) &= \inf\left\{y \in \mathbb{R} : F_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}; \varepsilon_d, \eta_d) \geq \tau\right\} \\
&= \inf\left\{y \in \mathbb{R} : T^{k_{d,c}}\left(F_{Y_d|R}(y \mid r_d^\star); \varepsilon_d, \eta_d\right) \geq \tau\right\} \\
&= \inf\left\{y \in \mathbb{R} : F_{Y_d|R}(y \mid r_d^\star) \geq \left(T^{k_{d,c}}\right)^{-1}(\tau, \epsilon_d, \eta_d)\right\} \\
&= Q_{Y_d|R}\left(\left(T^{k_{d,c}}\right)^{-1}(\tau, \epsilon_d, \eta_d) \mid r_d^\star\right).
\end{aligned}$$

By Assumption 3.2, $Q_{Y_d|R}(\cdot \mid r_d^\star)$ is continuous on $(0,1)$. By Lemma B.4, for each $\tau \in (0,1)$ and η_d, the map $\varepsilon_d \mapsto \left(T^{k_{d,c}}\right)^{-1}(\tau, \epsilon_d, \eta_d)$ is continuous on $[0,1]$, and therefore, the map $\varepsilon_d \mapsto Q_{Y_d|R}^{k_{d,c}}(\tau \mid r_{d'}; \varepsilon_d, \eta_d)$ is continuous on $[0,1]$.

Proof of Theorem 3.5

Proof. Let $d \in \{0,1\}$ and $r_{d'} \in \mathcal{R}_{d'}$. Recall that by Theorem 3.3, we have

$$\underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \leq F_{Y_d|R}(y \mid r_{d'}) \leq \overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}),$$

where

$$\underline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \equiv \max\left\{F_{Y_d|R}(y \mid r_d^\star) - k_{d,c}|r_{d'} - r_d^\star|, 0\right\},$$
$$\overline{F}_{Y_d|R}^{k_{d,c}}(y \mid r_{d'}) \equiv \min\left\{F_{Y_d|R}(y \mid r_d^\star) + k_{d,c}|r_{d'} - r_d^\star|, 1\right\}.$$

Thus, for any $\tau \in (0,1)$,

$$\left\{y : \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} \subseteq \left\{y : F_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} \subseteq \left\{y : \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\}.$$

Taking infima yields

$$\inf\left\{y : \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} \leq \inf\left\{y : F_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} = Q_{Y_d|R}(\tau|r_d) \leq \inf\left\{y : \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\}.$$

We first consider the lower bound. Note that

$$\begin{aligned}
\inf\left\{y : \overline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} &= \inf\left\{y : \min\left\{1, F_{Y_d|R}(y \mid r^\star_d) + \delta_d(r_{d'})\right\} \geq \tau\right\} \\
&= \inf\left\{y : F_{Y_d|R}(y \mid r^\star_d) + \delta_d(r_{d'}) \geq \tau\right\} \\
&= \inf\left\{y : F_{Y_d|R}(y \mid r^\star_d) \geq \tau - \delta_d(r_{d'})\right\},
\end{aligned}$$

where the second line follows from $\tau < 1$ and the fact that for any $a \in \mathbb{R}$, $\min\{1, a\} \geq \tau \iff (1 \geq \tau \text{and } a \geq \tau) \iff a \geq \tau$.

From the last equality, we evaluate the infimum as follows. If $\tau \leq \delta_d(r_{d'})$, then $\tau - \delta_d(r_{d'}) \leq 0$ and

$$\left\{y : F_{Y_d|R}(y \mid r^\star_d) \geq \tau - \delta_d\right\} = \left\{y : F_{Y_d|R}(y \mid r^\star_d) \geq 0\right\},$$

so the infimum equals $-\infty$. If $\tau > \delta_d(r_{d'})$, then $0 < \tau - \delta_d(r_{d'}) < 1$ and, by the definition of the quantile,

$$\inf\left\{y : F_{Y_d|R}(y \mid r^\star_d) \geq \tau - \delta_d(r_{d'})\right\} = Q_{Y|R,D=d}\left(\tau - \delta_d(r_{d'}) \mid r^\star_d\right).$$

Therefore, by the definition of $\underline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'})$,

$$\underline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'}) \leq Q_{Y_d|R}(\tau \mid r_{d'}).$$

We now consider the upper bound. Recall that

$$\begin{aligned}
\inf\left\{y : \underline{F}^{k_{d,c}}_{Y_d|R}(y \mid r_{d'}) \geq \tau\right\} &= \inf\left\{y : \max\left\{0, F_{Y_d|R}(y \mid r^\star_d) - \delta_d(r_{d'})\right\} \geq \tau\right\} \\
&= \inf\left\{y : F_{Y_d|R}(y \mid r^\star_d) - \delta_d(r_{d'}) \geq \tau\right\} \\
&= \inf\left\{y : F_{Y_d|R}(y \mid r^\star_d) \geq \tau + \delta_d(r_{d'})\right\},
\end{aligned}$$

where the second line follows from $\tau > 0$ and the fact that for any $a \in \mathbb{R}$, $\max\{0, a\} \geq \tau \iff$

$(0 \geq \tau)$ or $(a \geq \tau) \iff a \geq \tau$.

From the last equality, we evaluate the infimum as follows. If $\tau + \delta_d(r_{d'}) > 1$, then

$$\left\{y : y : F_{Y_d|R}(y \mid r_d^\star) \geq \tau + \delta_d(r_{d'})\right\} = \emptyset,$$

and thus the upper bound is $+\infty$. If $\tau + \delta_d(r_{d'}) \leq 1$, then $0 < \tau + \delta_d(r_{d'}) \leq 1$ and, by the definition of the quantile,

$$\inf\left\{y : F_{Y_d|R}(y \mid r_d^\star) \geq \tau + \delta_d(r_{d'})\right\} = Q_{Y|R,D=d}\left(\tau + \delta_d(r_{d'}) \mid r_d^\star\right).$$

Therefore, by the definition of $\overline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'})$,

$$Q_{Y_d|R}(\tau \mid r_{d'}) \leq \overline{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r_{d'}).$$

For $d \in \{0,1\}$, $r \in Supp(R)$, $\tau \in (0,1)$ and (ε_d, η_d), define the left-quantile of the extended cdf by

$$\widetilde{Q}^{k_{d,c}}_{Y_d|R}(\tau \mid r; \varepsilon_d, \eta_d) \equiv \inf\left\{y \in \mathbb{R} : \tilde{F}_{Y_d|R}\left(y \mid r; \varepsilon_d, \eta_d\right) \geq \tau\right\},$$

where $\tilde{F}_{Y_d|R}\left(y \mid r; \varepsilon_d, \eta_d\right)$ is defined in (14).

We distinguish two cases.

Case 1) $r \in \mathcal{R}_1$: By the definition of the extension,

$$\begin{aligned}
&\widetilde{Q}^{k_{1,c}}_{Y_1|R}(\tau \mid r; \varepsilon_1, \eta_1) - \widetilde{Q}^{k_{0,c}}_{Y_0|R}(\tau \mid r; \varepsilon_0, \eta_0) \\
=&\inf\left\{y \in \mathbb{R} : F_{Y|R,D=1}(y \mid r) \geq \tau\right\} - \inf\left\{y \in \mathbb{R} : F^{k_{0,c}}_{Y_0|R}(y \mid r; \varepsilon_0, \eta_0) \geq \tau\right\} \\
=&Q_{Y|R,D=1}(\tau \mid r) - Q^{k_{0,c}}_{Y_0|R}(\tau \mid r; \varepsilon_0, \eta_0) \qquad (16)
\end{aligned}$$

Let $e \in \left(\underline{\text{CQTE}}^{k_{0,c}}(\tau \mid r), \overline{\text{CQTE}}^{k_{0,c}}(\tau \mid r)\right)$. By the monotone approximation, there exists $\eta_0^\star > 0$ such that e is above the value of equation (16) evaluated at $(\varepsilon_0, \eta_0) = (0, \eta_0^\star)$ and below the value of the equation evaluated at $(\varepsilon_0, \eta_0) = (1, \eta_0^\star)$. Next, note that $Q^{k_{0,c}}_{Y_0|R}(\tau \mid r; \varepsilon_0, \eta_0)$ is continuous in ε_0 on $[0,1]$ by Lemma B.5. Thus, by the intermediate value theorem, there exists $\varepsilon_0^\star \in [0,1]$ such that equation (16) evaluated at $(\varepsilon_0, \eta_0) = (\varepsilon_0^\star, \eta_0^\star)$ yields e.

Case 2) $r \in \mathcal{R}_0$: By the definition of the extension,

$$\begin{aligned}
&\widetilde{Q}^{k_{1,c}}_{Y_1|R}(\tau \mid r; \varepsilon_1, \eta_1) - \widetilde{Q}^{k_{0,c}}_{Y_0|R}(\tau \mid r; \varepsilon_0, \eta_0) \\
=&\inf\left\{y \in \mathbb{R} : F^{k_{1,c}}_{Y_1|R}(y \mid r; \varepsilon_1, \eta_1) \geq \tau\right\} - \inf\left\{y \in \mathbb{R} : F_{Y|R,D=0}(y \mid r) \geq \tau\right\} \\
=&Q^{k_{1,c}}_{Y_1|R}(\tau \mid r; \varepsilon_1, \eta_1) - Q_{Y|R,D=0}(\tau \mid r). \qquad (17)
\end{aligned}$$

Let $e \in \left(\underline{\text{CQTE}}^{k_{1,c}}(\tau \mid r), \overline{\text{CQTE}}^{k_{1,c}}(\tau \mid r)\right)$. By the monotone approximation, there exists $\eta_1^\star > 0$ such that e is above the value of equation (17) evaluated at $(\varepsilon_1, \eta_1) = (1, \eta_1^\star)$ and below the value of the equation evaluated at $(\varepsilon_1, \eta_1) = (0, \eta_1^\star)$. Next, note that $Q^{k_{1,c}}_{Y_1|R}(\tau \mid r; \varepsilon_1, \eta_1)$ is continuous in ε_1 on $[0, 1]$ Lemma B.5. Thus, by the intermediate value theorem, there exists $\varepsilon_1^\star \in [0, 1]$ such that equation (17) evaluated at $(\varepsilon_1, \eta_1) = (\varepsilon_1^\star, \eta_1^\star)$ yields e.

In all, every value in the interior of $\left[\underline{\text{CQTE}}^{k_c}(\tau \mid r), \overline{\text{CQTE}}^{k_c}(\tau \mid r)\right]$ is attainable, so the interior of the identified set is sharp. ■

B.5 Proof of Theorem 4.1

Lemma B.6. *Suppose that Assumptions 4.1–4.6 hold. Then,*

$$\sqrt{n}\left(\hat{\beta}_d(\cdot) - \beta_d(\cdot)\right) \Rightarrow \mathbb{Z}_d(\cdot) \text{ in } l^\infty\left(\mathcal{T}_\beta\right)^{d_\beta},$$

where $\mathbb{Z}_d(\cdot)$ is a mean-zero Gaussian process with covariance kernel

$$Cov\left(\mathbb{Z}_d\left(\tau\right), \mathbb{Z}_d\left(\tilde{\tau}\right)\right) = \left(\frac{\min\left(\tau, \tilde{\tau}\right) - \tau\tilde{\tau}}{p_d}\right) J_d\left(\tau\right)^{-1} \Omega_d J_d\left(\tilde{\tau}\right)^{-1}.$$

Therefore,

$$\sqrt{n}\left[\mathcal{P}(r)'\hat{\beta}_d(\tau) - \mathcal{P}(r)'\beta_d(\tau)\right] \Rightarrow \mathcal{P}(r)'\mathbb{Z}_d(\tau)$$

in $l^\infty(\mathcal{T}_\beta \times \mathcal{R}_d)$.

Proof. Using the same argument as in Angrist et al. (2006), it is straightforward to see that

$$\sqrt{n_d}\left(\hat{\beta}_d(\cdot) - \beta_d(\cdot)\right) \Rightarrow \tilde{\mathbb{Z}}_d(\cdot) \text{in } l^\infty\left(\mathcal{T}_\beta\right)^{d_\beta},$$

where $\tilde{\mathbb{Z}}_d(\cdot)$ is a mean-zero Gaussian process with covariance kernel

$$Cov\left(\tilde{\mathbb{Z}}_d(\tau), \tilde{\mathbb{Z}}_d(\tilde{\tau})\right) = \left(\min\left(\tau, \tilde{\tau}\right) - \tau\tilde{\tau}\right) J_d\left(\tau\right)^{-1} \Omega_d J_d\left(\tilde{\tau}\right)^{-1}$$

under Assumptions 4.2–4.6. Since $n_d/n \to p_d$, we have

$$\sqrt{n}\left(\hat{\beta}_d(\cdot) - \beta_d(\cdot)\right) = \sqrt{\frac{n}{n_d}}\sqrt{n_d}\left(\hat{\beta}_d(\cdot) - \beta_d(\cdot)\right) \Rightarrow \mathbb{Z}_d(\cdot)$$

by the continuous mapping theorem.

Define the map

$$\Phi_d : l^\infty(\mathcal{T}_\beta)^{d_\beta} \to l^\infty(\mathcal{T}_\beta \times \mathcal{R}_d)$$

by $\Phi_d(b)(\tau, r) = \mathcal{P}(r)'b(\tau)$. Its Hadamard derivative at β_d in the direction h is $\Phi'_{d,\beta_d}(h)(\tau, r) = \mathcal{P}(r)'h(\tau)$. Since $\mathcal{P}(\cdot)$ is uniformly bounded on $\mathcal{R}_d$,

$$\|\Phi'_{d,\beta_d}(h)\|_\infty \leq \left(\sup_{r\in\mathcal{R}_d} \|\mathcal{P}(r)\|\right) \|h\|_\infty.$$

Thus, the Hadamard derivative is a continuous linear map. By the functional delta method, we obtain the second result. ■

For each $d \in \{0, 1\}$, a direct consequence of Lemma B.6 is that

$$\sqrt{n}\left[\begin{pmatrix} \widehat{\underline{Q}}_d^{k_{d,c}}(\tau|r) \\ \widehat{\overline{Q}}_d^{k_{d,c}}(\tau|r) \end{pmatrix} - \begin{pmatrix} \underline{Q}_d^{k_{d,c}}(\tau|r) \\ \overline{Q}_d^{k_{d,c}}(\tau|r) \end{pmatrix}\right] \Rightarrow \begin{pmatrix} \mathcal{P}(r_d^\star)'\mathbb{Z}_d\left(s_{d,L}(\tau, r; k_{d,c})\right) \\ \mathcal{P}(r_d^\star)'\mathbb{Z}_d\left(s_{d,U}(\tau, r; k_{d,c})\right) \end{pmatrix}$$

in $l^\infty(\mathcal{T}_{C,1-d} \times \mathcal{R}_{1-d})^2$.

Proof of the theorem

Proof. The joint weak convergence of $(\hat{\beta}_0, \hat{\beta}_1)$ follows from the same quantile regression argument by stacking the two processes. Lemma B.6 and the continuous mapping theorem then yield the result. ■

B.6 Proof of Theorem 4.2

Proof. Fix $d \in \{0, 1\}$ and let $j \in \{L, U\}$ index the lower or upper bound. For notational convenience, write $CQTE_{L,d} = \underline{CQTE}_d$, $CQTE_{U,d} = \overline{CQTE}_d$, $CATE_{L,d} = \underline{CATE}_d$, and $CATE_{U,d} = \overline{CATE}_d$, with the corresponding estimators defined analogously. On $\mathcal{T}_{C,d}$, Theorem 4.1 gives the weak convergence of the centered CQTE-bound estimator. On $\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}$, the counterfactual component is a known support endpoint and the only estimated component is the observed conditional quantile.

Hence, uniformly over $(\tau, r) \in (\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}) \times \mathcal{R}_d$,

$$\sqrt{n}\left\{\widehat{CQTE}_{j,d}(\tau, r) - CQTE_{j,d}(\tau, r)\right\} = (2d-1)\mathcal{P}(r)'\sqrt{n}\left\{\hat{\beta}_d(\tau) - \beta_d(\tau)\right\}.$$

For $\tau \in [0,1] \setminus \mathcal{T}_\beta$, both the population and estimated CQTE bounds use the same known support endpoints, so the centered difference is identically zero. Therefore, uniformly in $r \in \mathcal{R}_d$,

$$\begin{aligned}\sqrt{n}\left\{\widehat{CATE}_{j,d}(r) - CATE_{j,d}(r)\right\} = &\int_{\mathcal{T}_{C,d}} \sqrt{n}\left\{\widehat{CQTE}_{j,d}(\tau, r) - CQTE_{j,d}(\tau, r)\right\} d\tau \\ &+ (2d-1)\int_{\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}} \mathcal{P}(r)'\sqrt{n}\left\{\hat{\beta}_d(\tau) - \beta_d(\tau)\right\} d\tau.\end{aligned}$$

Theorem 4.1 and Lemma B.6, together with the joint weak convergence of $(\hat{\beta}_0, \hat{\beta}_1)$, imply the joint weak convergence of the two stochastic integrands. Define the map

$$\Psi_d : l^\infty(\mathcal{T}_{C,d} \times \mathcal{R}_d)^2 \times l^\infty(\mathcal{T}_\beta \times \mathcal{R}_d) \to l^\infty(\mathcal{R}_d)^2$$

by

$$\Psi_d(g_L, g_U, h)(r) \equiv \begin{pmatrix} \displaystyle\int_{\mathcal{T}_{C,d}} g_L(\tau, r)d\tau + (2d-1)\int_{\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}} h(\tau, r)d\tau \\ \displaystyle\int_{\mathcal{T}_{C,d}} g_U(\tau, r)d\tau + (2d-1)\int_{\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}} h(\tau, r)d\tau \end{pmatrix}.$$

Since the sets $\mathcal{T}_{C,d}$ and $\mathcal{T}_\beta \setminus \mathcal{T}_{C,d}$ are fixed, Ψ_d is a continuous linear map. The result therefore follows from the joint weak convergence of the CQTE-bound process and the quantile regression process and the continuous mapping theorem. ■

B.7 Proof of Theorem 4.3

Proof. Following Lemma D.1 of Chernozhukov et al. (2013), define the integration operator

$$\Phi(g_0, g_1, F) \equiv \sum_{d=0}^{1} \int_{\mathcal{R}_d} g_d(r)\, dF(r).$$

Then

$$\underline{ATE} = \Phi(\underline{CATE}_0, \underline{CATE}_1, F_R), \qquad \overline{ATE} = \Phi\big(\overline{CATE}_0, \overline{CATE}_1, F_R\big).$$

By Lemma D.1 of Chernozhukov et al. (2013), Φ is Hadamard differentiable. At (g_0, g_1, F_R), its derivative in the direction (h_0, h_1, H) is

$$\Phi'_{(g_0,g_1,F_R)}(h_0, h_1, H) = \sum_{d=0}^{1} \int_{\mathcal{R}_d} h_d(r)\, dF_R(r) + \sum_{d=0}^{1} \int_{\mathcal{R}_d} g_d(r)\, dH(r).$$

Define

$$h_L(r) \equiv \sum_{d=0}^{1} \mathbf{1}\{r \in \mathcal{R}_d\} \underline{CATE}_d(r), \qquad h_U(r) \equiv \sum_{d=0}^{1} \mathbf{1}\{r \in \mathcal{R}_d\} \overline{CATE}_d(r),$$

and

$$\mathbb{G}_{n,R,L} \equiv \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(h_L(R_i) - E[h_L(R)]\right), \qquad \mathbb{G}_{n,R,U} \equiv \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(h_U(R_i) - E[h_U(R)]\right).$$

Equivalently,

$$\mathbb{G}_{n,R,L} = \sqrt{n} \sum_{d=0}^{1} \left[\int_{\mathcal{R}_d} \underline{CATE}_d(r)\, d\left(\widehat{F}_R - F_R\right)(r) \right],$$

and

$$\mathbb{G}_{n,R,U} = \sqrt{n} \sum_{d=0}^{1} \left[\int_{\mathcal{R}_d} \overline{CATE}_d(r)\, d\left(\widehat{F}_R - F_R\right)(r) \right].$$

Let

$$\begin{pmatrix} \mathbb{G}_{R,L} \\ \mathbb{G}_{R,U} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_R \right), \qquad \Sigma_R \equiv \mathrm{Var}\left[\begin{pmatrix} h_L(R) \\ h_U(R) \end{pmatrix} \right].$$

Thus,

$$(\mathbb{G}_{n,R,L}, \mathbb{G}_{n,R,U}) \Rightarrow (\mathbb{G}_{R,L}, \mathbb{G}_{R,U})$$

by the multivariate central limit theorem.

We first establish

$$\left(\left\{ \sqrt{n}\left(\widehat{\beta}_d - \beta_d\right) \right\}_{d=0}^{1}, \mathbb{G}_{n,R,L}, \mathbb{G}_{n,R,U} \right) \Rightarrow \left(\{\mathbb{Z}_d\}_{d=0}^{1}, \mathbb{G}_{R,L}, \mathbb{G}_{R,U} \right). \tag{18}$$

By Theorem 5.1 and Step 1 of its proof in Appendix E.4 of Chernozhukov et al. (2013), the QR coefficient processes converge jointly with the empirical measure of R. Since h_L and h_U are fixed bounded functions of R, evaluating the empirical measure at these two functions yields the joint weak convergence in (18).

Since the CATE-bound processes are continuous transformations of the QR coefficient processes,

the continuous mapping theorem yields

$$\left(\left\{\sqrt{n}\left(\widehat{\underline{CATE}}_d - \underline{CATE}_d\right)\right\}_{d=0}^{1}, \left\{\sqrt{n}\left(\widehat{\overline{CATE}}_d - \overline{CATE}_d\right)\right\}_{d=0}^{1}, \mathbb{G}_{n,R,L}, \mathbb{G}_{n,R,U}\right)$$
$$\Rightarrow \left(\{\mathbb{Z}_{C,L,d}\}_{d=0}^{1}, \{\mathbb{Z}_{C,U,d}\}_{d=0}^{1}, \mathbb{G}_{R,L}, \mathbb{G}_{R,U}\right).$$

Applying the functional delta method to Φ at $(\underline{CATE}_0, \underline{CATE}_1, F_R)$ gives

$$\sqrt{n}\left(\widehat{\underline{ATE}} - \underline{ATE}\right) = \sum_{d=0}^{1}\int_{\mathcal{R}_d} \sqrt{n}\left\{\widehat{\underline{CATE}}_d(r) - \underline{CATE}_d(r)\right\} dF_R(r)$$
$$+ \mathbb{G}_{n,R,L} + o_p(1).$$

Similarly,

$$\sqrt{n}\left(\widehat{\overline{ATE}} - \overline{ATE}\right) = \sum_{d=0}^{1}\int_{\mathcal{R}_d} \sqrt{n}\left\{\widehat{\overline{CATE}}_d(r) - \overline{CATE}_d(r)\right\} dF_R(r)$$
$$+ \mathbb{G}_{n,R,U} + o_p(1).$$

Finally, since integration with respect to the fixed distribution F_R is a continuous linear map, the continuous mapping theorem and the joint weak convergence above imply

$$\sqrt{n}\left(\widehat{\underline{ATE}} - \underline{ATE}\right) \Rightarrow \sum_{d=0}^{1}\int_{\mathcal{R}_d} \mathbb{Z}_{C,L,d}(r)\, dF_R(r) + \mathbb{G}_{R,L},$$

and

$$\sqrt{n}\left(\widehat{\overline{ATE}} - \overline{ATE}\right) \Rightarrow \sum_{d=0}^{1}\int_{\mathcal{R}_d} \mathbb{Z}_{C,U,d}(r)\, dF_R(r) + \mathbb{G}_{R,U}.$$

■

C Estimation and Inference for the Bounds on the Distribution Functions

We employ the distributional regression (DR) considered by Foresi and Peracchi (1995) and Chernozhukov et al. (2013) to estimate the distribution functions of the potential outcomes. Let $d \in \{0,1\}$ be given. We then assume that for each $y \in \mathcal{Y}_d$ and $r \in \mathcal{R}_d$,

$$F_{Y_d|R}(y|r) = \Lambda\left(\mathcal{P}(r)'\beta_d(y)\right), \tag{19}$$

where $\mathcal{P}(r)$ is a vector of transformations of r such as polynomials or B-splines, and $\Lambda(\cdot)$ is a known link function such as the logistic or standard normal distribution function. Note that the semiparametric specification of the conditional distribution in (19) is imposed only for $r \in \mathcal{R}_d$ and need not hold for $r \in \mathcal{R}_{1-d}$.

Let $\hat{\beta}_d(y)$ denote an estimator of $\beta_d(y)$. It can be obtained by solving the following optimization problem:

$$\hat{\beta}_d(y) = \arg\max_{b \in \mathbb{R}^{d_\beta}} \sum_{i \in \mathcal{J}_d} \left[\mathbf{1}(Y_i \leq y) \log\left(\Lambda\left(\mathcal{P}(R_i)'b\right)\right) + \mathbf{1}(Y_i > y) \log\left(1 - \Lambda\left(\mathcal{P}(R_i)'b\right)\right)\right]$$

for each $y \in \mathcal{Y}_d$. Then, the DR estimator of $F_{Y_d|R}$ is

$$\hat{F}_{Y_d|R}(y|r) \equiv \Lambda\left(\mathcal{P}(r)'\hat{\beta}_d(y)\right).$$

Once we obtain an estimator of $F_{Y_d|R}(\cdot|\cdot)$ on $\mathcal{R}_d$, the conditional counterfactual distribution bounds in (2) can be estimated by

$$\begin{aligned}
\widehat{\underline{F}}^{k_{d,c}}_{Y_d|R}(y|r) &\equiv \max\left\{\hat{F}_{Y_d|R}(y|r_d^\star) - k_{d,c}|r - r_d^\star|, 0\right\}, \\
\widehat{\overline{F}}^{k_{d,c}}_{Y_d|R}(y|r) &\equiv \min\left\{\hat{F}_{Y_d|R}(y|r_d^\star) + k_{d,c}|r - r_d^\star|, 1\right\},
\end{aligned}$$

for $r \in \mathcal{R}_{1-d}$. Corresponding estimators of the lower and upper bounds on the unconditional distribution of Y_d are then

$$\begin{aligned}
\widehat{\underline{F}}^{k_{d,c}}_{Y_d}(y) &\equiv \frac{1}{n}\sum_{i=1}^{n}\left[\mathbf{1}\{R_i \in \mathcal{R}_d\}\hat{F}_{Y_d|R}(y|R_i) + \mathbf{1}\{R_i \in \mathcal{R}_{1-d}\}\widehat{\underline{F}}^{k_{d,c}}_{Y_d|R}(y|R_i)\right], \\
\widehat{\overline{F}}^{k_{d,c}}_{Y_d}(y) &\equiv \frac{1}{n}\sum_{i=1}^{n}\left[\mathbf{1}\{R_i \in \mathcal{R}_d\}\hat{F}_{Y_d|R}(y|R_i) + \mathbf{1}\{R_i \in \mathcal{R}_{1-d}\}\widehat{\overline{F}}^{k_{d,c}}_{Y_d|R}(y|R_i)\right].
\end{aligned}$$

These estimators are the sample analogues of the bounds in Corollary 3.4 and average the observed and counterfactual-bound components over the full empirical distribution of R.

The standard weak convergence theory for the DR estimators can be found in Chernozhukov et al. (2013). Unlike the bounds on the conditional quantile functions, the bounds on the conditional distribution functions involve the max and min operators. Since these operators are Hadamard directionally differentiable, the limiting distributions of the estimated conditional bounds can be obtained using the delta method for Hadamard directionally differentiable functions; see Fang and

Santos (2019). Once weak convergence of the conditional bound estimators is established, the unconditional bounds can be handled using the same integration-operator argument as in the proof of Theorem 4.3.

D Identification of the Means of Counterfactual Outcomes

In this section, we investigate the identification of the mean of counterfactual outcomes without considering identification of the distribution functions. We assume that $E[|Y_d|] < \infty$ and define $g_d(r) \equiv E[Y_d|R = r]$.

Assumption D.1. *The following conditions hold:*

(i) For each $d \in \{0,1\}$, $E[Y_d|R = r_d]$ is identified for any $r_d \in \mathcal{R}_d$.

(ii) Let $r, \tilde{r} \in Supp(R)$ be given. There exist $k_{0,m}, k_{1,m} \geq 0$ such that

$$\left| E[Y_1|R = r] - E[Y_1|R = \tilde{r}] \right| \leq k_{1,m} \left| r - \tilde{r} \right|$$

and

$$\left| E[Y_0|R = r] - E[Y_0|R = \tilde{r}] \right| \leq k_{0,m} \left| r - \tilde{r} \right|.$$

Assumption D.1-(i) can be viewed as a special case of Assumption 2.1. Specifically, Assumption D.1-(i) holds when Assumption 2.1 is satisfied with $h(x) = x$. Assumption D.1-(ii) imposes a global Lipschitz restriction on $g_d(r)$. The following theorem formalizes the corresponding extrapolation results.

Theorem D.1. *Suppose that Assumption D.1 holds. For any given $r_d \in \mathcal{R}_d$ with $d \in \{0,1\}$, we have*

$$E[Y_1|R = r_0] \in [LB_{1,m}(r_0, k_{1,m}), UB_{1,m}(r_0, k_{1,m})],$$
$$E[Y_0|R = r_1] \in [LB_{0,m}(r_1, k_{0,m}), UB_{0,m}(r_1, k_{0,m})],$$

where

$$
\begin{aligned}
LB_{1,m}(k_{1,m}, r_0) &= E\left[Y_1 | R = r_1^\star\right] - k_{1,m} \cdot \left| r_0 - r_1^\star \right|, \\
UB_{1,m}(k_{1,m}, r_0) &= E\left[Y_1 | R = r_1^\star\right] + k_{1,m} \cdot \left| r_0 - r_1^\star \right|, \\
LB_{0,m}(k_{0,m} r_1) &= E\left[Y_0 | R = r_0^\star\right] - k_{0,m} \cdot \left| r_1 - r_0^\star \right|, \\
UB_{0,m}(k_{0,m}, r_1) &= E\left[Y_0 | R = r_0^\star\right] + k_{0,m} \cdot \left| r_1 - r_0^\star \right|.
\end{aligned}
$$

Moreover, these bounds are sharp.

Proof. We only prove the sharpness of the bounds with $LB_{1,m}$ and $UB_{1,m}$, and the sharpness of $LB_{0,m}$ and $UB_{0,m}$ can be shown by a similar way.

To verify the sharpness of $UB_{1,m}(k_{1,m}, r_0)$, consider the following DGP:

$$
\tilde{g_1}(r) = \begin{cases} g_1(r) & \text{if } r \in \mathcal{R}_1, \\ \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E[Y \mid R = \tilde{r}] + k_{1,m} \cdot \left| r - \tilde{r} \right| \right\} & \text{if } r \in \mathcal{R}_0. \end{cases}
$$

This construction ensures that $\tilde{g_1}(r_0)$ attains the upper bound $UB_{1,m}(k_{1,m}, r_0)$ for every $r_0 \in \mathcal{R}_0$. Note that if $r \in \mathcal{R}_1$, then by construction, $\tilde{g_1}(r) = g_1(r)$, which is identified over $\mathcal{R}_1$. Thus Assumption D.1-(i) holds with this DGP. We now verify that this DGP satisfies Assumption D.1-(ii); that is, for any $r_1, r_2 \in Supp(R)$ such that $r_1 < r_2$, we have $|\tilde{g_1}(r_2) - \tilde{g_1}(r_1)| \leq k_{1,m}|r_2 - r_1|$. We consider the following three cases:

Case 1) $r_1, r_2 \in \mathcal{R}_1$: Since $\tilde{g_1}(r_j) = g_1(r_j)$ for $j = 1, 2$, and $g_1(\cdot)$ satisfies Assumption D.1-(ii), it is obvious that

$$
\begin{aligned}
|\tilde{g_1}(r_2) - \tilde{g_1}(r_1)| &= |g_1(r_2) - g_1(r_1)| \\
&\leq k_{1,m}|r_2 - r_1|.
\end{aligned}
$$

Case 2) $r_1 \in \mathcal{R}_0$, $r_2 \in \mathcal{R}_1$: We have

$$
\begin{aligned}
\tilde{g_1}(r_1) &= \inf_{\tilde{r} \in \mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m} \left| r_1 - \tilde{r} \right| \right\} \\
&\leq E\left[Y_1 \mid R = r_2\right] + k_{1,m} \left| r_1 - r_2 \right| \\
&= \tilde{g_1}(r_2) + k_{1,m} \left| r_1 - r_2 \right|,
\end{aligned}
$$

and therefore,

$$\tilde{g_1}(r_1) - \tilde{g_1}(r_2) \leq k_{1,m}\Big|r_1 - r_2\Big|.$$

Conversely, since g_1 satisfies Assumption D.1-(ii), for all $\tilde{r} \in \mathcal{R}_1$,

$$g_1(r_2) \leq g_1(\tilde{r}) + k_1|r_2 - \tilde{r}|,$$

so

$$\begin{aligned}
\tilde{g_1}(r_2) &= g_1(r_2) \\
&\leq \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ g_1(\tilde{r}) + k_{1,m}\Big|r_2 - \tilde{r}\Big| \right\} \\
&\leq \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ g_1(\tilde{r}) + k_{1,m}\Big|r_1 - \tilde{r}\Big| \right\} + k_{1,m}\Big|r_2 - r_1\Big| \\
&= \tilde{g_1}(r_1) + k_{1,m}\Big|r_1 - r_2\Big|,
\end{aligned}$$

which yields that

$$|\tilde{g_1}(r_2) - \tilde{g_1}(r_1)| \leq k_{1,m}\Big|r_1 - r_2\Big|.$$

Case 3): $r_1, r_2 \in \mathcal{R}_0$: By definition,

$$\tilde{g_1}(r_j) = \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m}\Big|r_j - \tilde{r}\Big| \right\}$$

for each $j = 1, 2$. Then, by the triangle inequality,

$$\begin{aligned}
&|\tilde{g_1}(r_2) - \tilde{g_1}(r_1)| \\
=&\left| \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m}\Big|r_2 - \tilde{r}\Big| \right\} - \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m}\Big|r_1 - \tilde{r}\Big| \right\} \right| \\
\leq&\left| \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m}\Big|r_1 - \tilde{r}\Big| \right\} + k_{1,m}\Big|r_2 - r_1\Big| - \inf_{\tilde{r}\in\mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] + k_{1,m}\Big|r_1 - \tilde{r}\Big| \right\} \right| \\
=&k_{1,m}\Big|r_2 - r_1\Big|.
\end{aligned}$$

In all, Assumption D.1-(ii) holds with the DGP.

We now consider the sharpness of the lower bound on $g_1(r_0)$. Consider the following DGP:

$$\bar{g_1}(r) = \begin{cases} g_1(r) & \text{if } r \in \mathcal{R}_1, \\ \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] - k_{1,m} \left| r - \tilde{r} \right| \right\} & \text{if } r \in \mathcal{R}_0. \end{cases}$$

This construction ensures that $\bar{g_1}(r_0)$ attains the lower bound $LB_{1,m(}(k_1, r_0)$ for every $r_0 \in \mathcal{R}_0$. Note that if $r \in \mathcal{R}_1$, then by construction, $\bar{g_1}(r) = g_1(r)$, and thus Assumption D.1-(i) holds with this DGP.

We now verify that this DGP satisfies Assumption D.1-(ii); that is, for any $r_1 < r_2$, $|\bar{g_1}(r_2) - \bar{g_1}(r_1)| \leq k_{1,m}|r_2 - r_1|$. We consider the following three cases:

Case 1) $r_1, r_2 \in \mathcal{R}_1$: Since $\bar{g_1}(r_j) = g_1(r_j)$ for $j = 1, 2$ and $g_1(\cdot)$ satisfies Assumption D.1-(ii), it follows that

$$\begin{aligned} \left| \bar{g_1}(r_2) - \bar{g_1}(r_1) \right| &= \left| g_1(r_2) - g_1(r_1) \right| \\ &\leq k_{1,m} \left| r_2 - r_1 \right|. \end{aligned}$$

Case 2) $r_1 \in \mathcal{R}_0$ and $r_2 \in \mathcal{R}_1$: By the definition of $\bar{g_1}$, we have

$$\begin{aligned} \bar{g_1}(r_1) &= \sup_{\tilde{r} \in \mathcal{R}_1} \left\{ E\left[Y_1 \mid R = \tilde{r}\right] - k_{1,m} \left| r_1 - \tilde{r} \right| \right\} \\ &\geq E\left[Y_1 | R = r_2\right] - k_{1,m} \left| r_1 - r_2 \right| \\ &= \bar{g_1}(r_2) - k_{1,m}|r_2 - r_1|, \end{aligned}$$

and therefore,

$$\bar{g_1}(r_1) - \bar{g_1}(r_2) \geq -k_{1,m}|r_2 - r_1|.$$

Conversely, since g_1 satisfies Assumption D.1-(ii), it follows that, for all $\tilde{r} \in \mathcal{R}_1$,

$$g_1(\tilde{r}) \leq g_1(r_2) + k_{1,m} \left| r_2 - \tilde{r} \right|,$$

which is equivalent to $g_1(\tilde{r}) - k_{1,m} \left| r_2 - \tilde{r} \right| \leq g_1(r_2)$.Putting them together, we have

So

$$\begin{aligned}\bar{g_1}(r_2) &= g_1(r_2) \\ &\geq \sup_{\tilde{r}\in\mathcal{R}_1}\left\{g_1(\tilde{r}) - k_{1,m}\left|r_2-\tilde{r}\right|\right\} \\ &\geq \sup_{\tilde{r}\in\mathcal{R}_1}\left\{g_1(\tilde{r}) - k_{1,m}|r_1-\tilde{r}|\right\} - k_{1,m}\left|r_2-r_1\right| \\ &= \bar{g_1} - k_{1,m}\left|r_2-r_1\right|.\end{aligned}$$

As a result, we obtain $\left|\bar{g_1}(r_2)-\bar{g_1}(r_1)\right| \leq k_{1,m}\left|r_2-r_1\right|$ in this case.

Case 3) $r_1, r_2 \in \mathcal{R}_0$: By construction,

$$\bar{g_1}(r_j) = \sup_{\tilde{r}\in\mathcal{R}_1}\left\{E\left[Y_1 \mid R=\tilde{r}\right] - k_{1,m}\left|r_j-\tilde{r}\right|\right\}$$

for each $j=1,2$. Then, by the triangle inequality, we have

$$\begin{aligned}&\left|\bar{g_1}(r_2)-\bar{g_1}(r_1)\right| \\ =&\left|\sup_{\tilde{r}\in\mathcal{R}_1}\left\{E\left[Y_1|R=\tilde{r}\right] - k_{1,m}\left|r_2-\tilde{r}\right|\right\} - \sup_{\tilde{r}\in\mathcal{R}_1}\left\{E\left[Y_1|R=\tilde{r}\right] - k_{1,m}\left|r_1-\tilde{r}\right|\right\}\right| \\ \leq&\left|\sup_{\tilde{r}\in\mathcal{R}_1}\left\{E\left[Y_1|R=\tilde{r}\right] - k_{1,m}\left|r_1-\tilde{r}\right|\right\} + k_{1,m}\left|r_2-r_1\right| - \sup_{\tilde{r}\in\mathcal{R}_1}\left\{E\left[Y_1|R=\tilde{r}\right] - k_{1,m}\left|r_1-\tilde{r}\right|\right\}\right| \\ =&k_{1,m}|r_2-r_1|,\end{aligned}$$

implying that Assumption Assumption D.1-(ii) holds with this DGP. Therefore, the bounds $LB_{1,m}(k_{1,m}, r_0)$ and $UB_{1,m}(k_{1,m}, r_0)$ are sharp. We can show the sharpness of $LB_{0,m}(k_{0,m}, r_1)$ and $UB_{0,m}(k_{0,m}, r_1)$ by using a similar argument. ■

Theorem D.1 establishes two important results. First, it characterizes the identified set for the conditional expectation function of the potential outcome on the running variable over regions in which point identification may fail. Second, it shows that the derived bounds are sharp. The results in Theorem D.1 is useful for constructing bounds on the unconditional means of Y_1 and Y_0, and thus, allows us to construct bounds on the average treatment effect for the whole population. Specifically, suppose that we are interested in identification of $E\left[Y_1 - Y_0|R=r_1\right]$ for $r_1 \in \mathcal{R}_1 - \{c\}$. Note that $E\left[Y_1|R=r_1\right] = E\left[Y|R=r_1\right]$ is identified from the data. On the other hand, Theorem D.1 provides the sharp bounds on $E\left[Y_0|R=r_1\right]$, and the bounds are given by $[LB_{0,m}(k_{0,m}, r_1), UB_{0,m}(k_{0,m}, r_1)]$.

As a result, it is straightforward to see that

$$E\left[Y_1 - Y_0 | R = r_1\right] \in \left[E\left[Y_1|R = r_1\right] - UB_{0,m}(k_{0,m}, r_1), E\left[Y_1|R = r_1\right] - LB_{0,m}(k_{0,m}, r_1)\right].$$

The same argument can be applied pointwise to each value of the running variable in its support. Integrating the resulting bounds with respect to the distribution of the running variable then yields bounds on the average treatment effect. The following corollary formalizes this result.

Corollary D.2. *Suppose that Assumption D.1 holds and that* $\mathcal{R}_1 \cup \mathcal{R}_0 = Supp(R)$ *with* $\Pr\left(\mathcal{R}_1 \cap \mathcal{R}_0\right) = 0$. *Let* $\mathcal{C}$ *be a measurable subset of* $Supp(R)$ *and* $ATE(\mathcal{C}) \equiv E\left[Y_1 - Y_0 | R \in \mathcal{C}\right]$. *Then, we have*

$$ATE(\mathcal{C}) \in \left[LB^{ATE}\left(k_{0,m}, k_{1,m}; \mathcal{C}\right), UB^{ATE}\left(k_{0,m}, k_{1,m}; \mathcal{C}\right)\right],$$

where

$$\begin{aligned}
LB^{ATE}\left(k_{0,m}, k_{1,m}; \mathcal{C}\right) \equiv & \int_{\mathcal{R}_0 \cap \mathcal{C}} LB_{1,m}(k_{1,m}, r) - E\left[Y|R = r\right] dF_R(r) \\
& + \int_{\mathcal{R}_1 \cap \mathcal{C}} E\left[Y|R = r\right] - UB_{0,m}(k_{0,m}, r) dF_R(r), \\
UB^{ATE}\left(k_{0,m}, k_{1,m}; \mathcal{C}\right) \equiv & \int_{\mathcal{R}_1 \cap \mathcal{C}} E\left[Y|R = r\right] - LB_{0,m}(k_{0,m}, r) dF_R(r) \\
& + \int_{\mathcal{R}_0 \cap \mathcal{C}} UB_{1,m}(k_{1,m}, r) - E\left[Y|R = r\right] dF_R(r).
\end{aligned}$$

When $\mathcal{C} = Supp(R)$, then the resulting bounds are for the average treatment effect for the whole population. We note that the bounds in Corollary D.2 need not be sharp. To obtain sharp bounds for certain causal parameters, we instead rely on identification of the distributions of the potential outcomes.

The following assumption can be considered as a version of SMRV assumption (Assumption 3.3) for the counterfactual mean functions, which we call the "monotone running variable" assumption.

Assumption D.2 (Monotone Running Variable (MRV))**.** *For any* $r, \tilde{r} \in Supp(R)$ *such that* $r \geq \tilde{r}$, *we have*

$$E\left[Y_d|R = r\right] \geq E\left[Y_d|R = \tilde{r}\right]$$

for each $d \in \{0, 1\}$.

Assumption D.2 posits that the average potential outcome associated with a higher value of the

running variable is not smaller than that associated with a lower value. This assumption is closely related to the MIV assumption in Manski and Pepper (2000) and Manski and Pepper (2009).

Under Assumptions D.1 and D.2, we obtain the following bounds on the conditional expectation functions of the potential outcomes:

$$E\left[Y_1|R=r_0\right] \in \left[LB_1^{MRV}(k_{1,m}, r_0), UB_1^{MRV}(k_{1,m}, r_0)\right],$$
$$E\left[Y_0|R=r_1\right] \in \left[LB_0^{MRV}(k_{0,m}, r_1), UB_0^{MRV}(k_{1,m}, r_1)\right],$$

where

$$\begin{aligned}
LB_1^{MRV}(k_{1,m}, r_0) &\equiv E\left[Y_1|R=r_1^\star\right] - k_{1,m} \cdot \left|r_0 - r_1^\star\right|, \\
UB_1^{MRV}(k_{1,m}, r_0) &\equiv E\left[Y_1|R=r_1^\star\right], \\
LB_0^{MRV}(k_{0,m}, r_1) &\equiv E\left[Y_0|R=r_0^\star\right], \\
UB_0^{MRV}(k_{0,m}, r_1) &\equiv E\left[Y_0|R=r_0^\star\right] + k_{0,m} \cdot \left|r_0^\star - r_1\right|,
\end{aligned}$$

where $r_d \in \mathcal{R}_d$ for each $d \in \{0,1\}$.

The above bounds are similar to those in Theorem D.1. However, $UB_1^{MRV}(k_{1,m}, r_0)$ and $LB_0^{MRV}(k_{0,m}, r_1)$ are tighter than their respective counterparts. This tightening follows from the fact that Assumption D.2 imposes a one-sided monotonicity restriction on the conditional mean function of the potential outcomes with respect to the running variable, which is consistent with the results established in the literature.